

Fine-mapping and colocalization

DCEG Statistical Genetics Workshop

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Agenda

Background

Purpose of post-GWAS analysis

Fine mapping

Single and multiple effects, summary data, annotations, and ancestry

Implementation

Preparing a locus, choosing LD, and checking the result

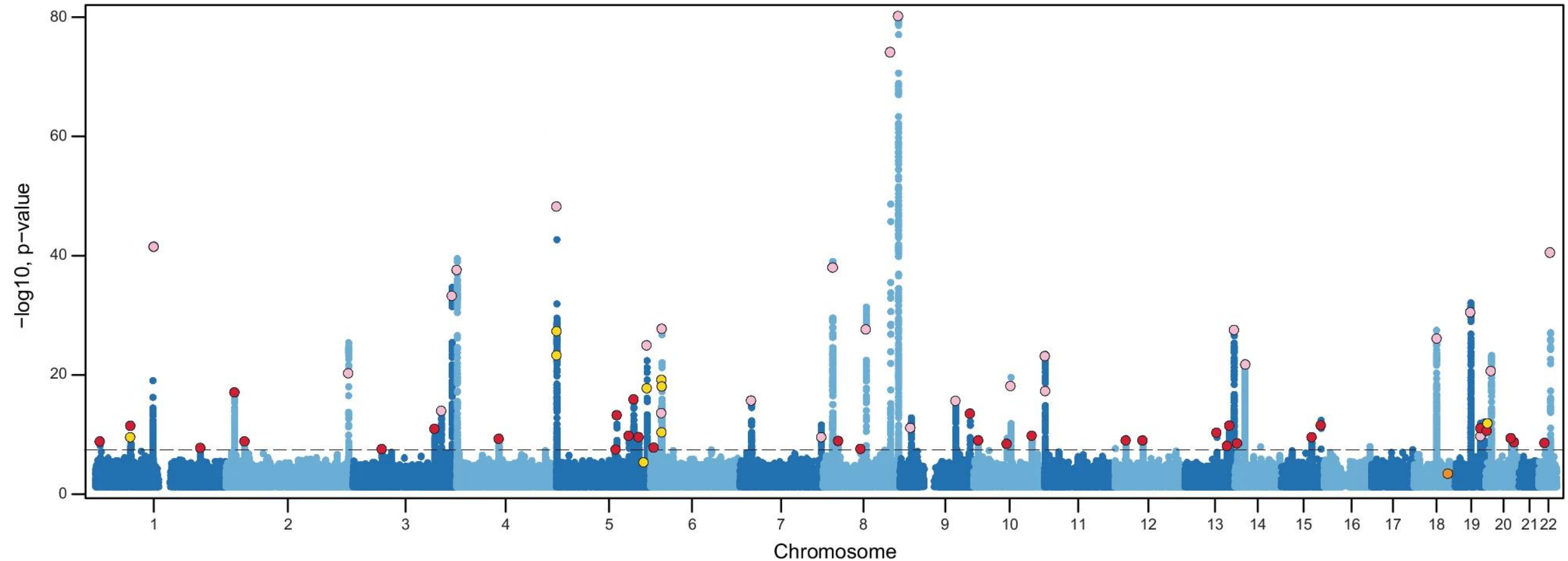
Colocalization

Shared variants across traits and careful interpretation

GWAS Catalog in July 2026

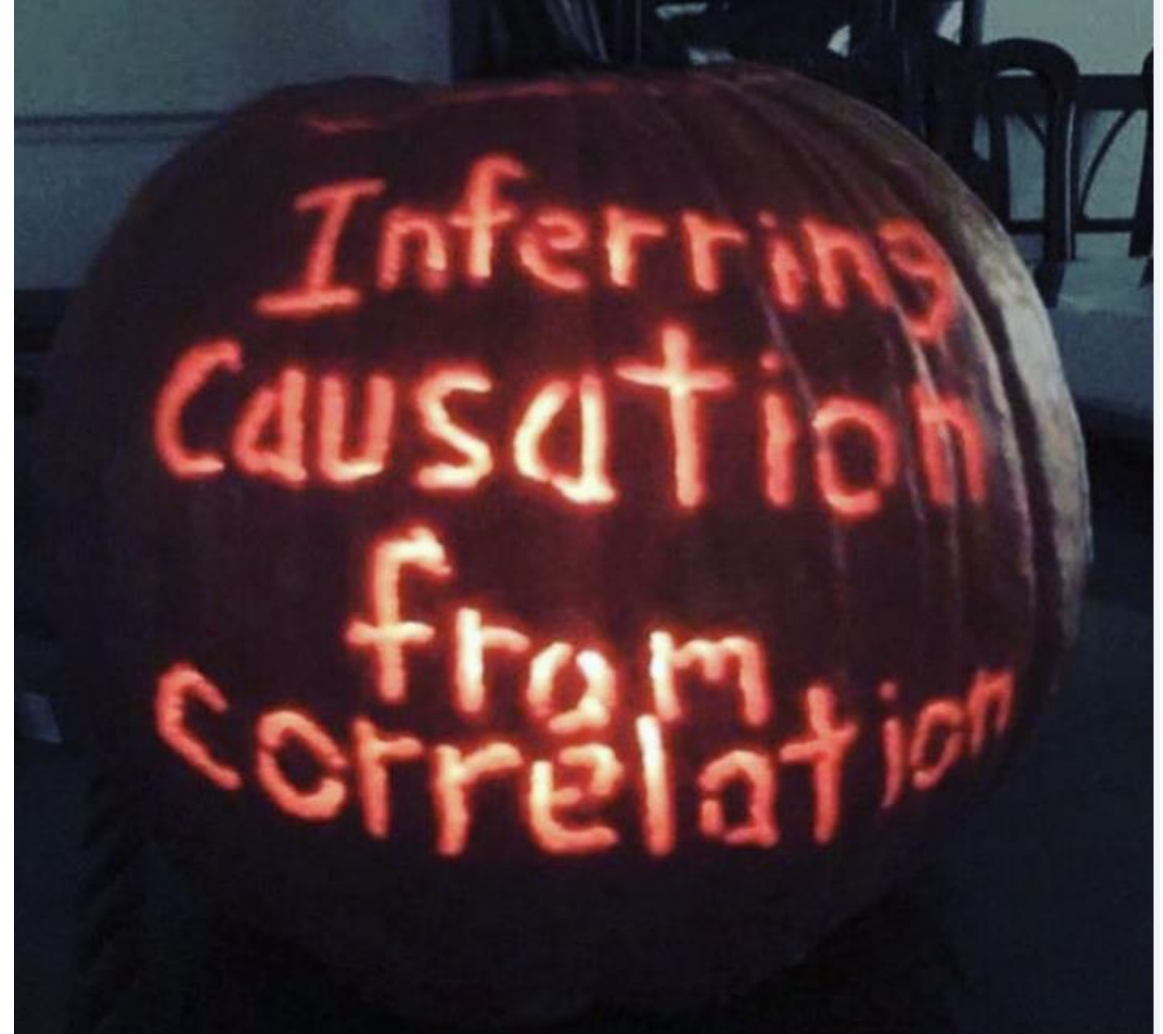


Example GWAS: identifying risk loci



Interpreting a GWAS association

- GWAS **identifies associations** not biological mechanism
 - Cannot tell us how the variants affect the outcome
- Does not imply the causality of associated genetic variants
- Need to identify exact genetic variants with “**true effects**” from associations arising due to high correlation with the true signals



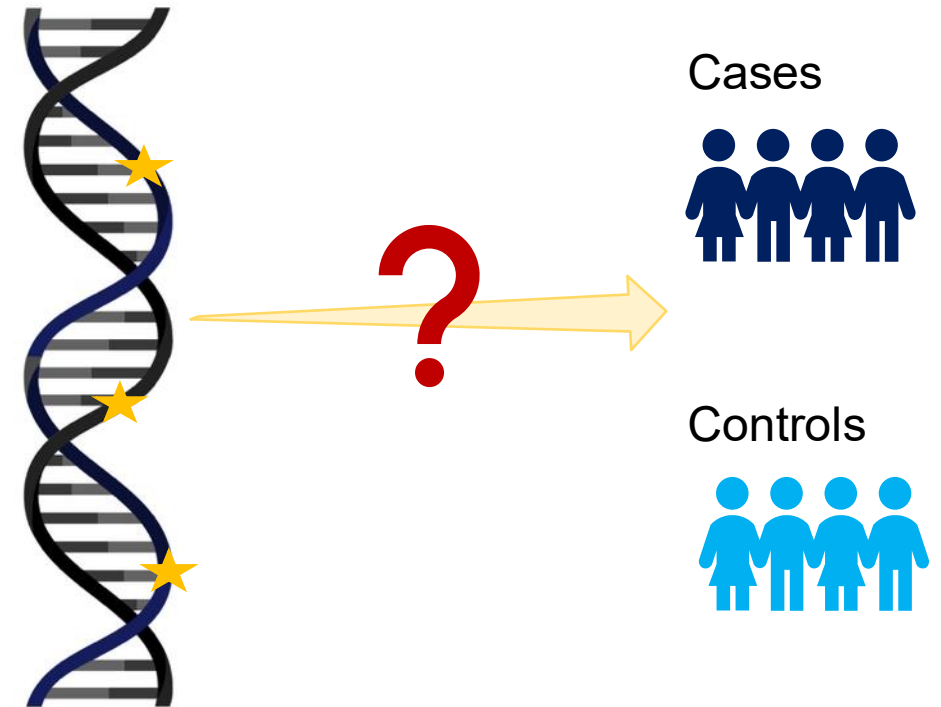
Molecular mechanisms after GWAS

Many GWAS associations are in the **non-coding region** of the genome

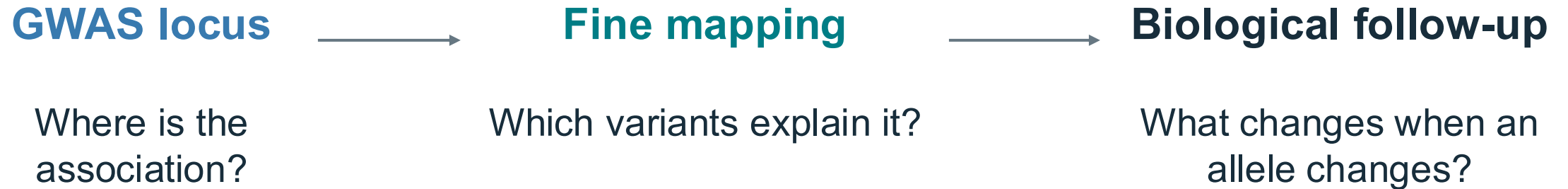
Mechanism remains largely **unknown**

Possibly acts by **regulating molecular phenotypes** like gene expression, protein levels, metabolites.

A direction of **post-GWAS analysis**: Identifying **intermediate molecular phenotypes** through which genetic risk may be mediated, by **integrating relevant molecular data**



Fine mapping in the post-GWAS workflow



TODAY

Variant probabilities and credible sets
Colocalization with molecular traits

LATER SESSIONS

Integrative omics, TWAS / PWAS, gene and tissue prioritization, pathway analysis, and experimental mechanism

Statistical and biological causality

- A variant is **statistically causal** for the outcome if the coefficient of the SNP in the underlying joint model is non-zero
- A SNP is **biologically causal** for the outcome if the SNP influences phenotypic variation (or case control status) through a biological mechanism
- Fine-mapping addresses **statistical causality**
- The purpose of ***in silico* post-GWAS analyses** and identifying statistical causality is to propose a smaller number of candidate variants to be investigated downstream

Statistical and biological causality

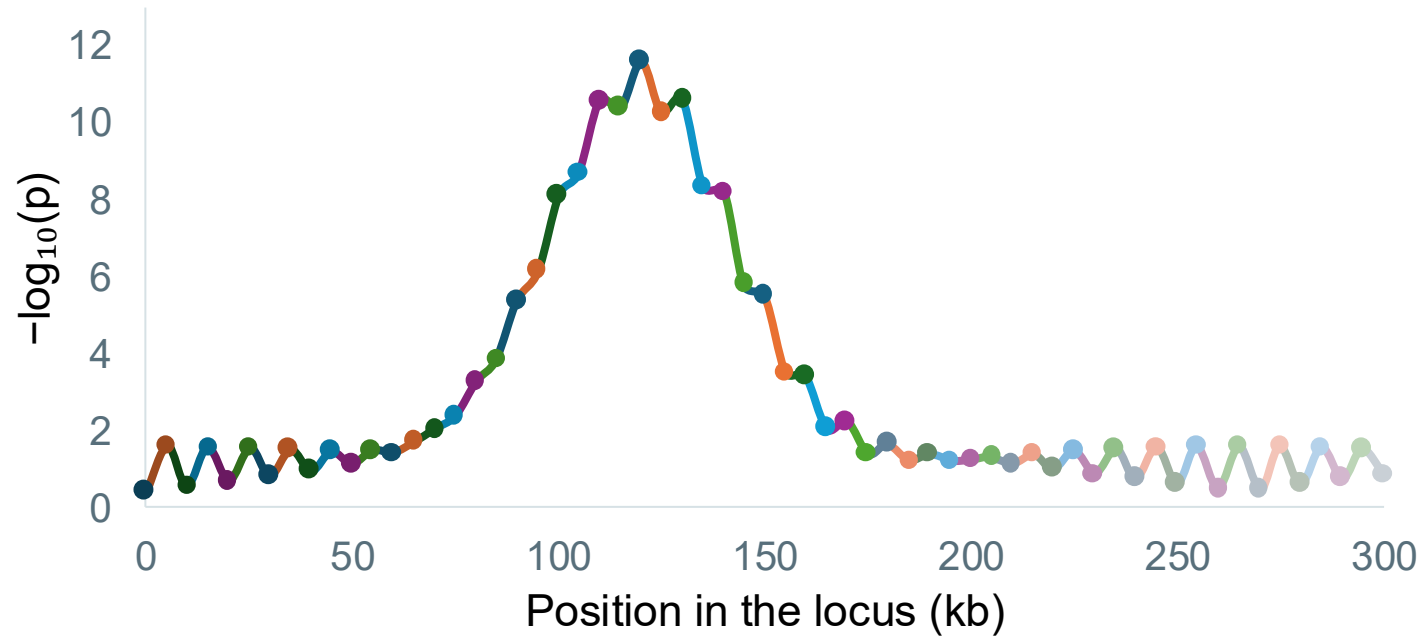
	Statistical causality	Biological causality
Question	Which variants have non-zero joint effects under the model?	What changes when the allele changes?
Evidence	GWAS association signal and LD, Optional: genomic characteristics	Perturbation and molecular evidence in a relevant context
Conclusion	Candidate variants and uncertainty	A mechanism linking allele to phenotype

A non-zero joint effect prioritizes a candidate. Biological mechanisms need further evidence.

What fine mapping enables

- Find candidate causal variants
causal regulatory and coding variations
- Prioritize regulatory mechanisms
Suggest perturbation and knock out experiments
- Understand genetic architecture
Enrichment of genetic features
Similarities and differences across diseases or ancestry-groups
Prediction

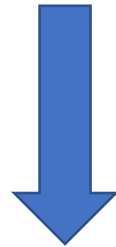
A GWAS locus can contain many associated variants



- How does this GWAS association pattern arise
- Hypothesis: simple underlying causal structure
- Which SNP(s) is(are) causal?

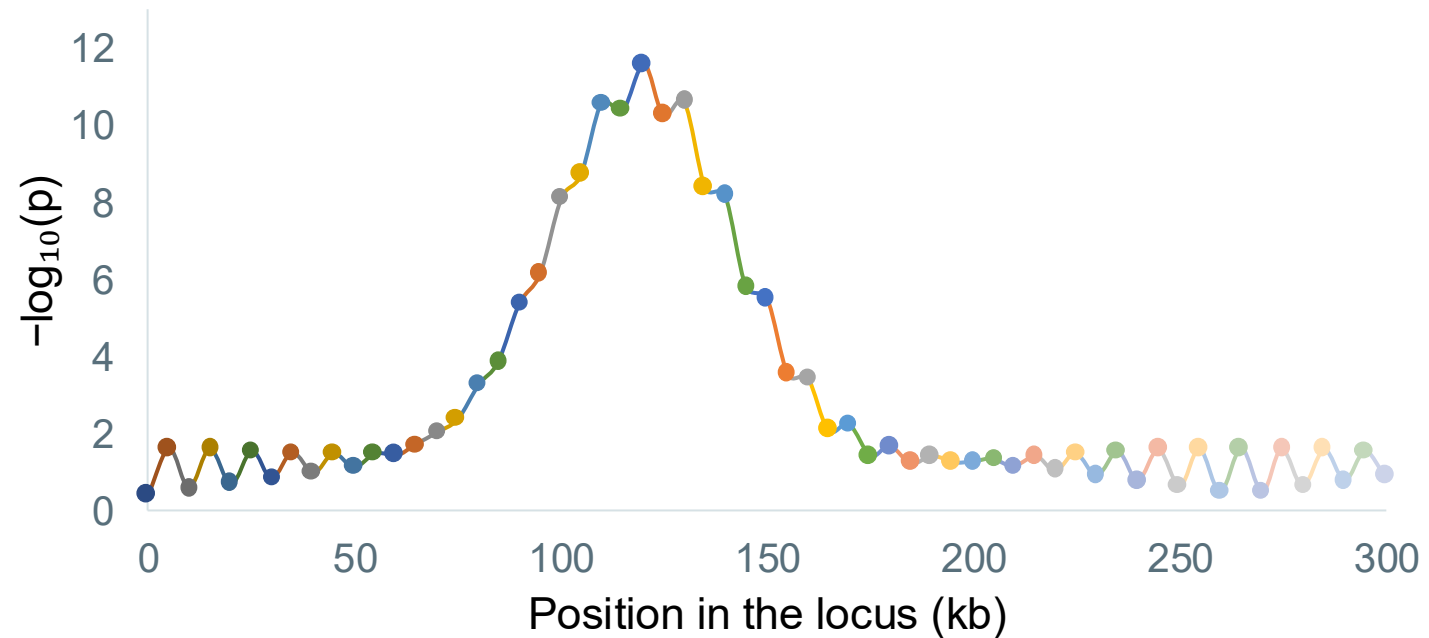
Underlying effects and observed associations

True Causal
Status



- Correlation between SNPs (LD)
- Sampling noise/error

Observed
association
patterns in GWAS



Linkage disequilibrium creates ambiguity

Variants inherited together can tag the same signal

Sample	SNP A	SNP B	SNP C	SNP D
Sample 1	G	T	A	C
Sample 2	G	T	A	T
Sample 3	A	C	G	C
Sample 4	A	C	G	T
Sample 5	G	T	A	C
Sample 6	A	C	G	T

A, B, and C are in perfect LD in this toy example.

Their marginal associations will be identical here (apart from allele coding).

Marginal vs Joint models

GWAS is Marginal

Estimate effects of variants one by one

Is the variant associated with the outcome?

A correlated proxy can predict very well.

Fine-mapping is Joint

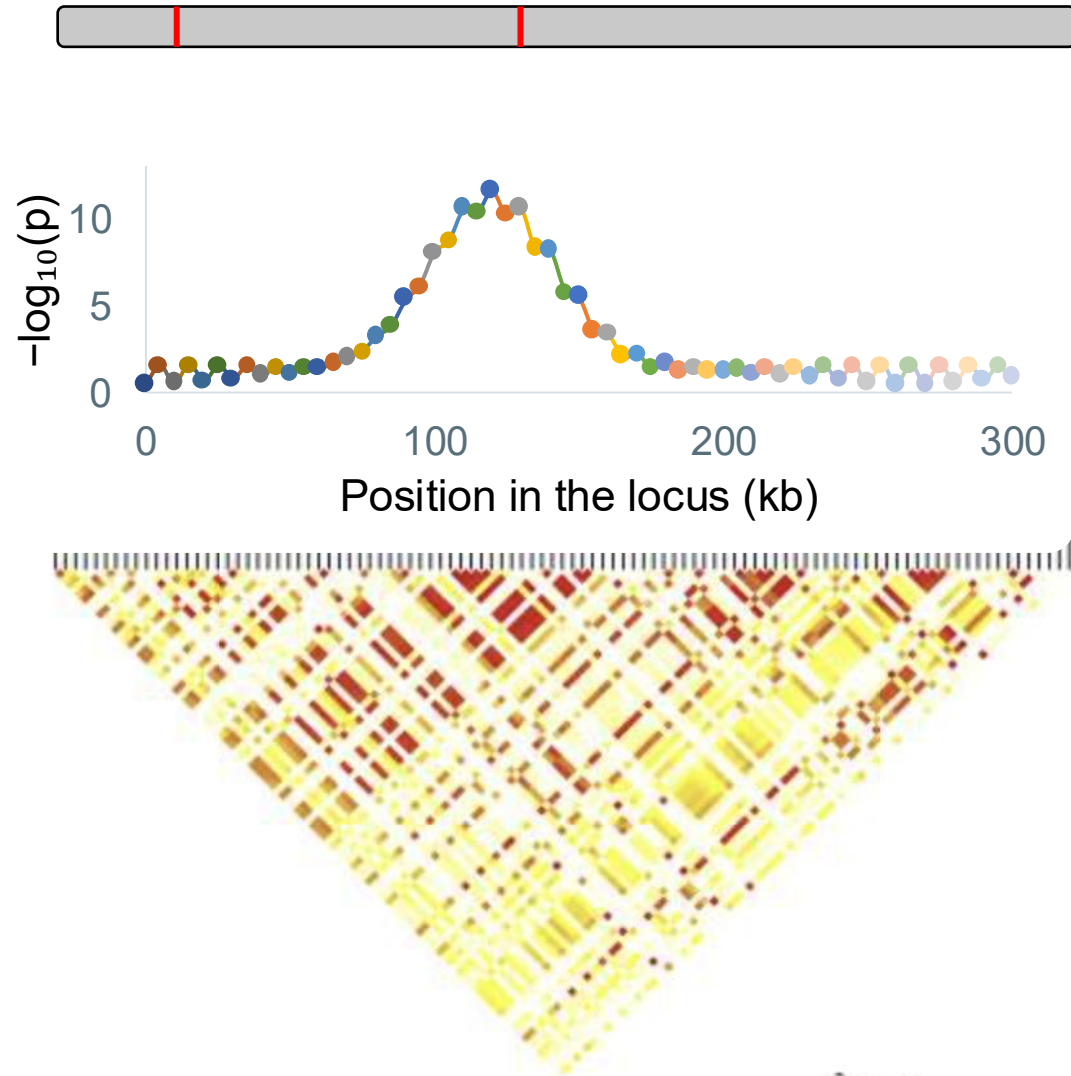
Estimate effects of all variants in a locus together

Which variants have non-zero after accounting for the other variants?

Fine-mapping also represents uncertainty about the answer.

The lead SNP can tag another variant

True Causal
Status



SNPs in high LD (correlated) with the true causal variant(s) can potentially be significant marginally

Often the SNP with the lowest p-value in a locus (called **sentinel/index SNP**) is in high LD (correlated) with the true causal SNP(s)

Sentinel SNPs might not be causal but “tags” the true causal SNP(s)

The true causal SNP(s) is possibly in a set of SNPs that are in LD with the sentinel SNP

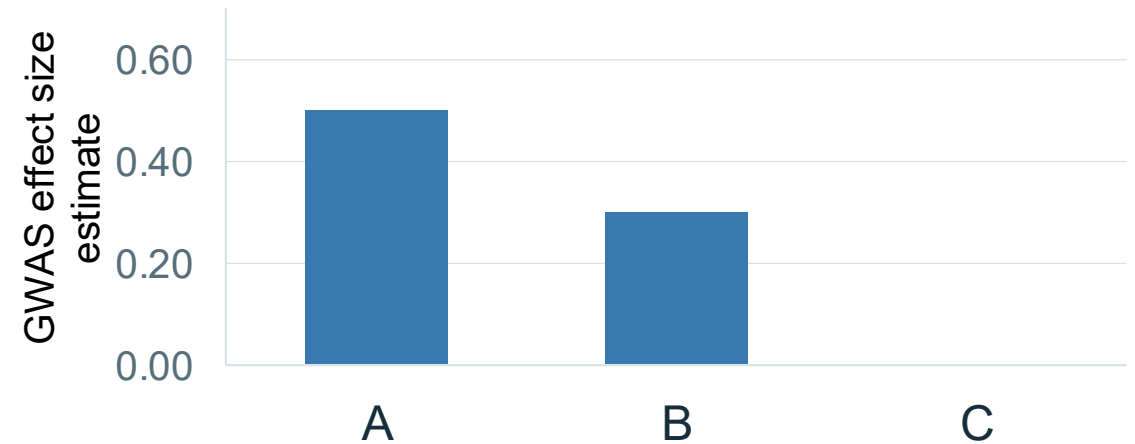
Scenario 1: one causal variant, several associations

SNP A has a true effect. B is in LD with A. C is uncorrelated with A.

Underlying effects



Expected marginal GWAS effects



Association at B does not imply a non-zero joint effect at B.

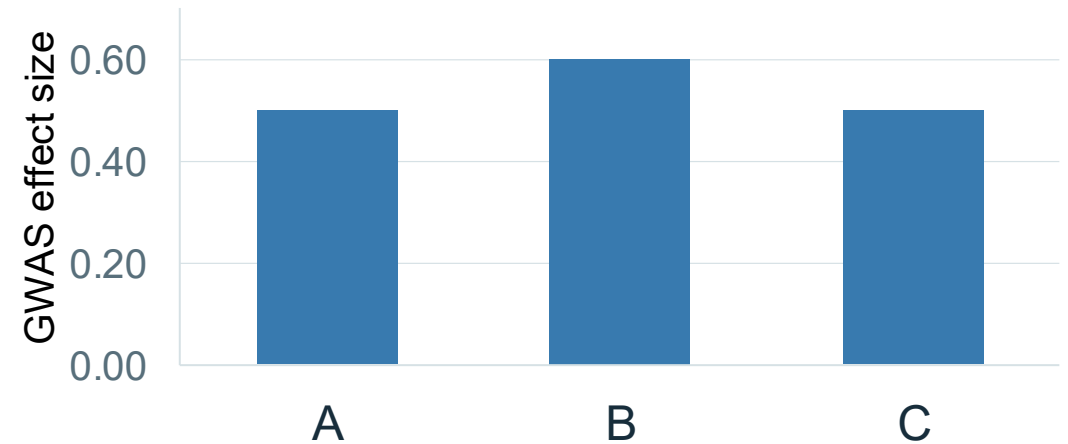
Scenario 2: a tag can have the largest association

A and C both have true effect. B is correlated with both. A and C are independent

Underlying effects



Expected marginal GWAS effects

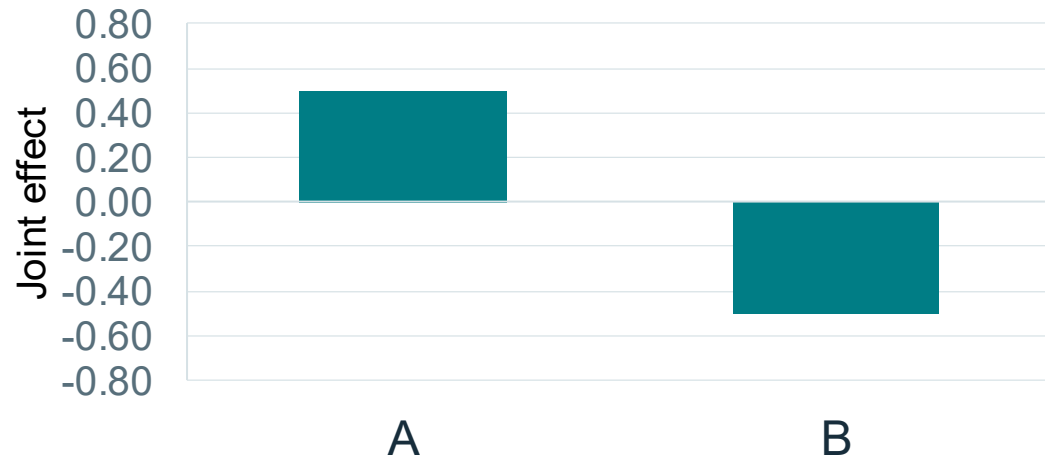


B tags both effects, even though its own joint effect is zero.

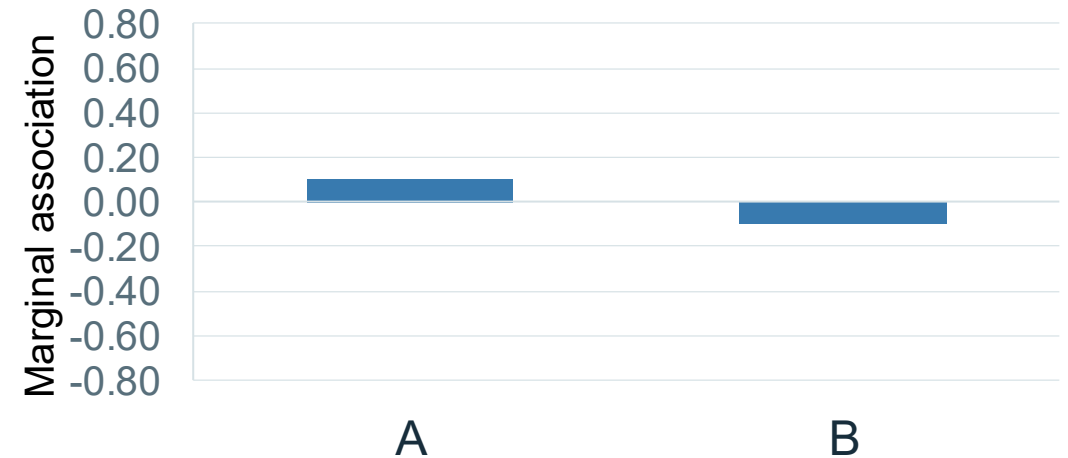
Scenario 3: opposite effects can mask each other

Genotypes A and B are positively correlated, but their effects have opposite signs

Underlying effects



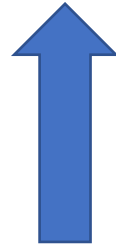
Expected marginal GWAS effects



A small marginal effect can hide a larger joint effect.

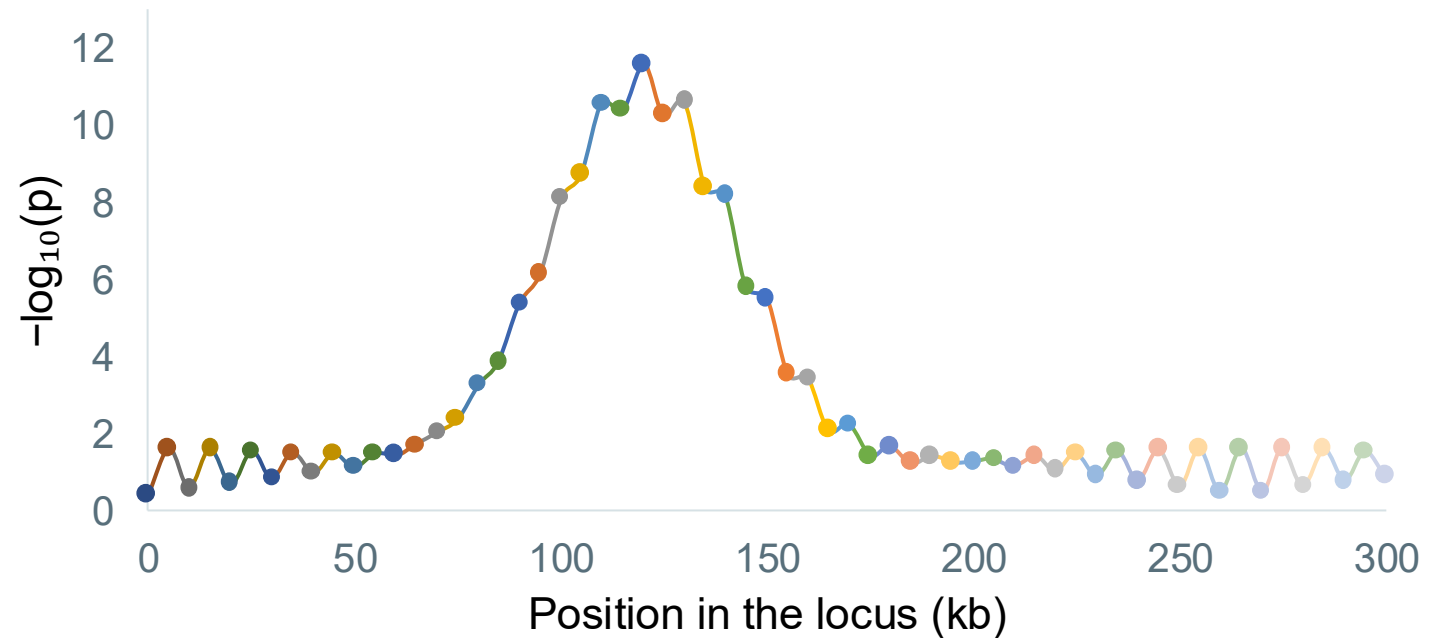
The goal of fine mapping

True Causal
Status



Infer which SNP(s) are causal in the locus

Observed
association
patterns in GWAS



The fine-mapping agenda for today

Inference

Identifying (with uncertainty) candidate causal variants

Association and LD

Why summary statistics can reconstruct the joint-model evidence

Current models

Multiple effects, functional priors, and multiple ancestries

Practical analysis

Inputs, LD sources, diagnostics, and interpretation

The quantities we want to estimate

- Probability that a variant is statistically causal
- A small set of variants that contains a causal variant with a given probability

Fine mapping with one causal variant

One signal
Several possible candidate causal variants

Bayesian Methods

Posterior inclusion probabilities and credible sets

PIP (Posterior Inclusion Probability)

$P(\text{Non-zero effect} \mid \text{data, model})$: probability of a non-zero joint effect

95% credible set

A set carrying at least 95% posterior probability for an effect

Within a set interpretation

Variants are interpreted as candidate causal variants

How to compute: Bayes Factor

BF_{10} = evidence under an effect model / evidence under the null

$$ABF = \sqrt{\frac{V+W}{V}} \exp\left(-\frac{z^2}{2} \frac{W}{(V+W)}\right)$$

Z The signed GWAS test statistic

V Estimated variance of the estimated effect

W Prior variance for plausible non-zero effects

One causal variant: evidence becomes probability

Exactly one causal variant among the analyzed SNPs

$$PIP_j = \frac{BF_j}{\sum_k BF_k}$$

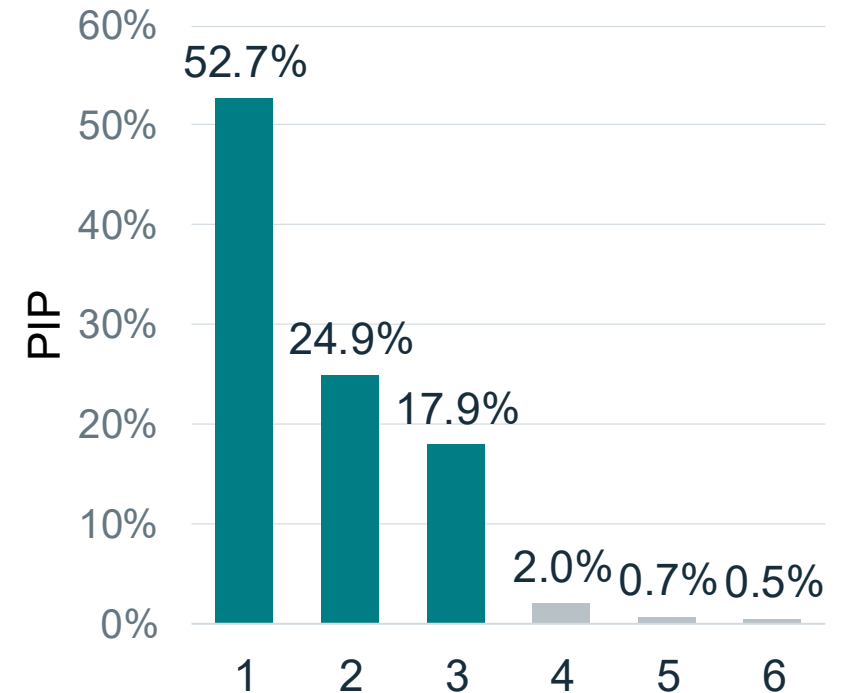
Candidate	Bayes factor	Equal-prior PIP
A	10	10 / 21 = 0.476
B	10	10 / 21 = 0.476
C	1	1 / 21 = 0.048

A + B = 0.952 → 95% credible set = {A, B}

A 95% credible set, SNP by SNP

To construct **smallest 95% CS**: Order PIPs in decreasing order and cumulatively add them until the sum of PIPs is ≥ 0.95

SNP	GWAS P value	PIP	Cumulative
rs2912781	9.39E-22	0.52705	52.705%
rs10736303	1.96E-21	0.24881	77.586%
rs7895676	2.79E-21	0.17922	95.508%
rs12356902	8.10E-05	0.02005	97.513%
rs146141333	2.07E-03	0.00675	98.188%
rs2912778	7.96E-21	0.00495	98.683%

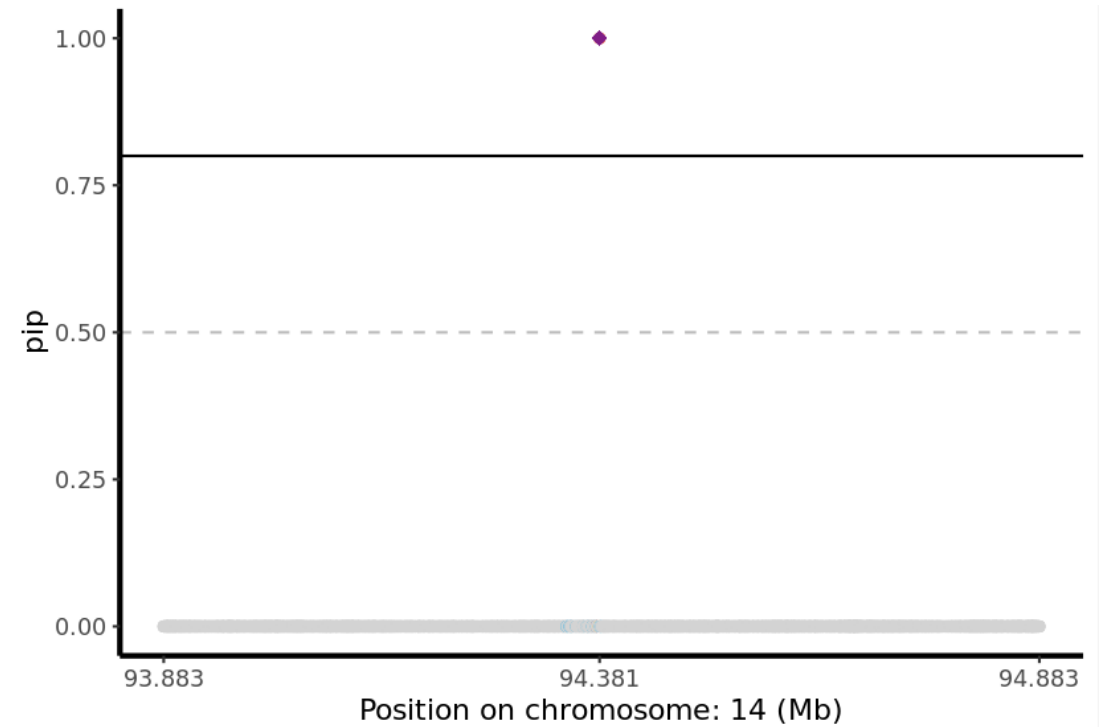
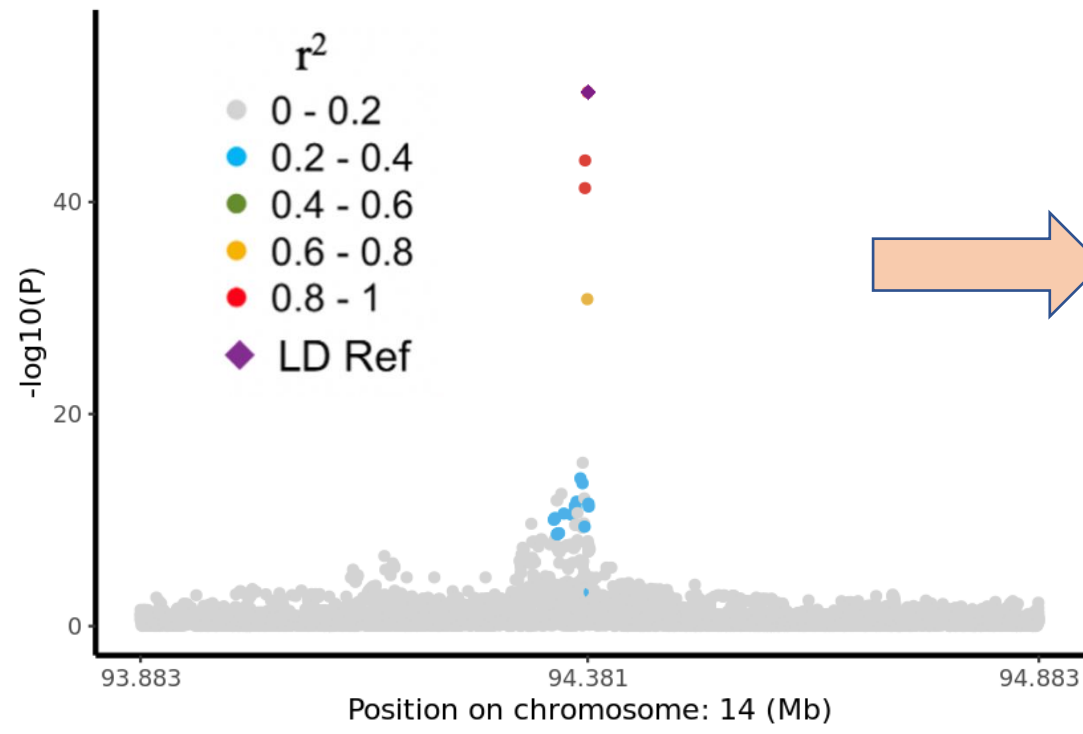


First 3 SNPs carry 95.508% probability

Outside the set $\neq$ zero probability

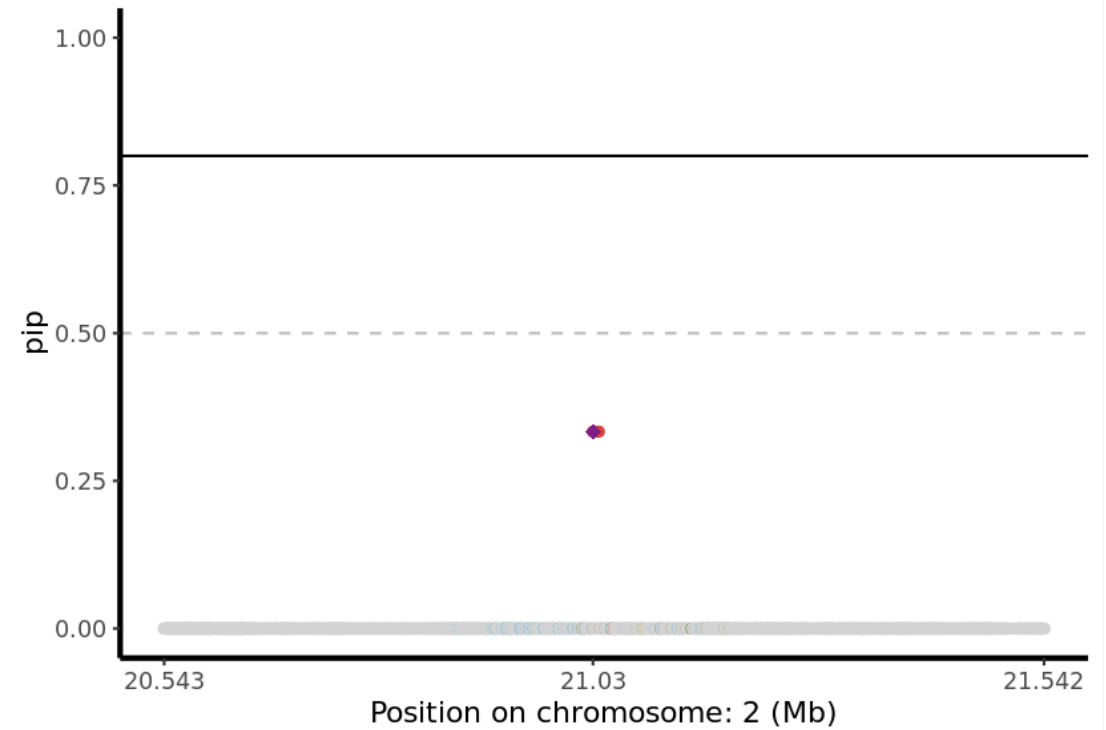
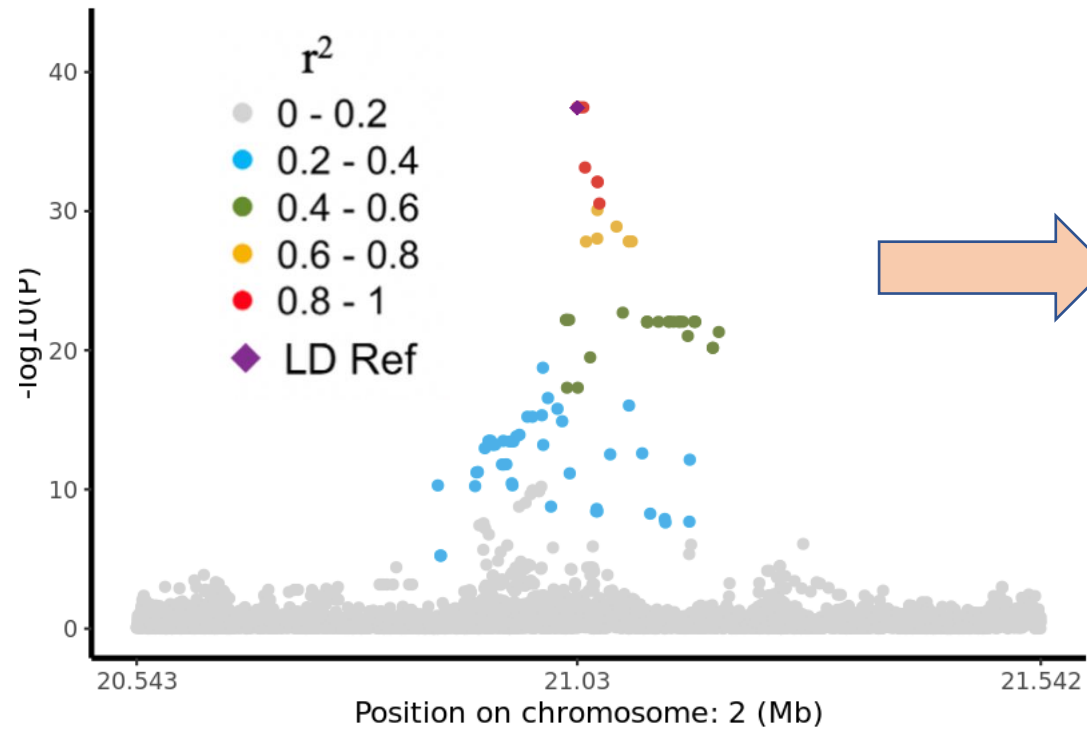
A singleton credible set

Clear difference in GWAS association even among high LD SNPs



A credible set with several candidates

Minor differences in GWAS association among high LD SNPs



Pop quiz 1: what belongs in the credible set?

One causal variant. PIPs: $A = 0.60$, $B = 0.36$, $C = 0.04$.

What is the smallest 95% credible set?

- A. $\{A\}$
- B. $\{A, B\}$
- C. $\{A, B, C\}$

Why summary statistics can support fine mapping

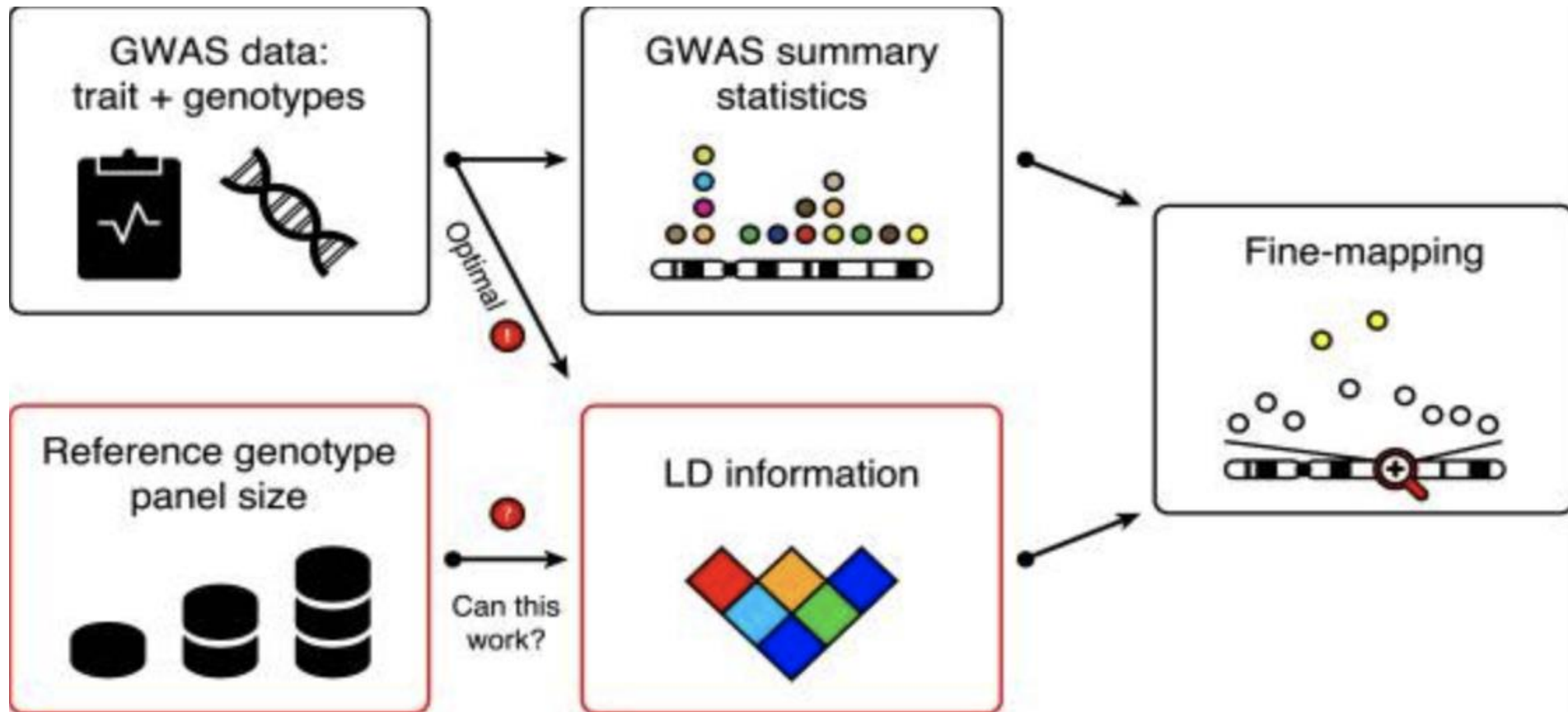


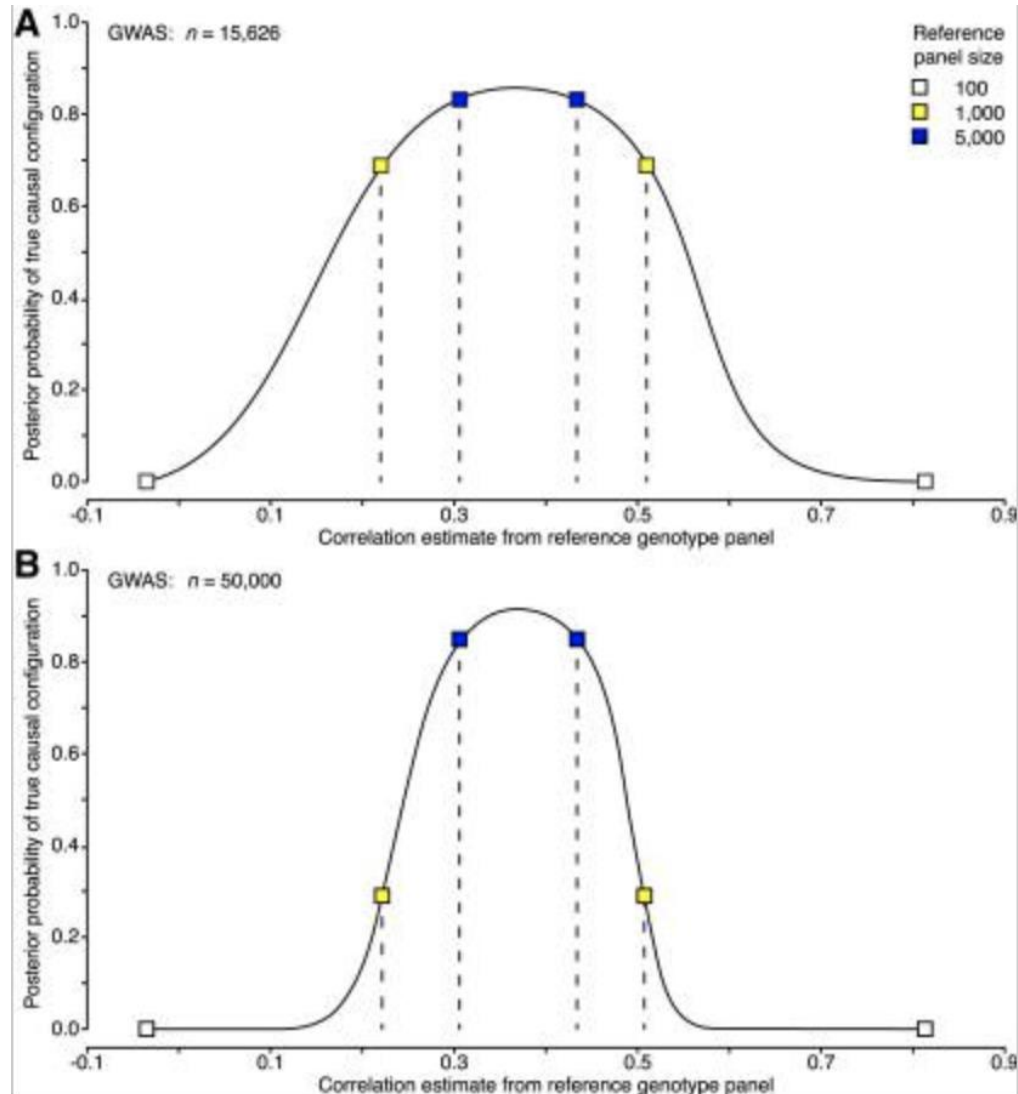
Image from: Benner et al (2017); *AJHG*.

Fine-mapping inputs and outputs

Inputs, in identical SNP order	Outputs and interpretation
Effects and SE, or signed z scores Sample size and trait type Build, position, effect / other allele	PIP for each variant Probability of a non-zero joint effect
Signed LD correlation matrix R Matched population and allele coding Allele frequency / QC as required	Credible sets for modeled effects Alternative variants within each set
Individual-level option Genotypes, phenotype, covariates	Model diagnostics Convergence, CS purity, sensitivity

Summary-data methods use association evidence together with LD.

LD reference-panel size and ancestry match



Reference panel size must scale with the sample size of GWAS

Populations must **match closely in terms of ancestral background**

Simulation with 2 variants: 1 causal & 1 non-causal with correlation 0.37

Image from: Benner et al (2017); *AJHG*.

Imputation and study size matters

Earlier GWAS: 1000G imputation

Later GWAS: HRC imputation

Identifies “refined” credible sets

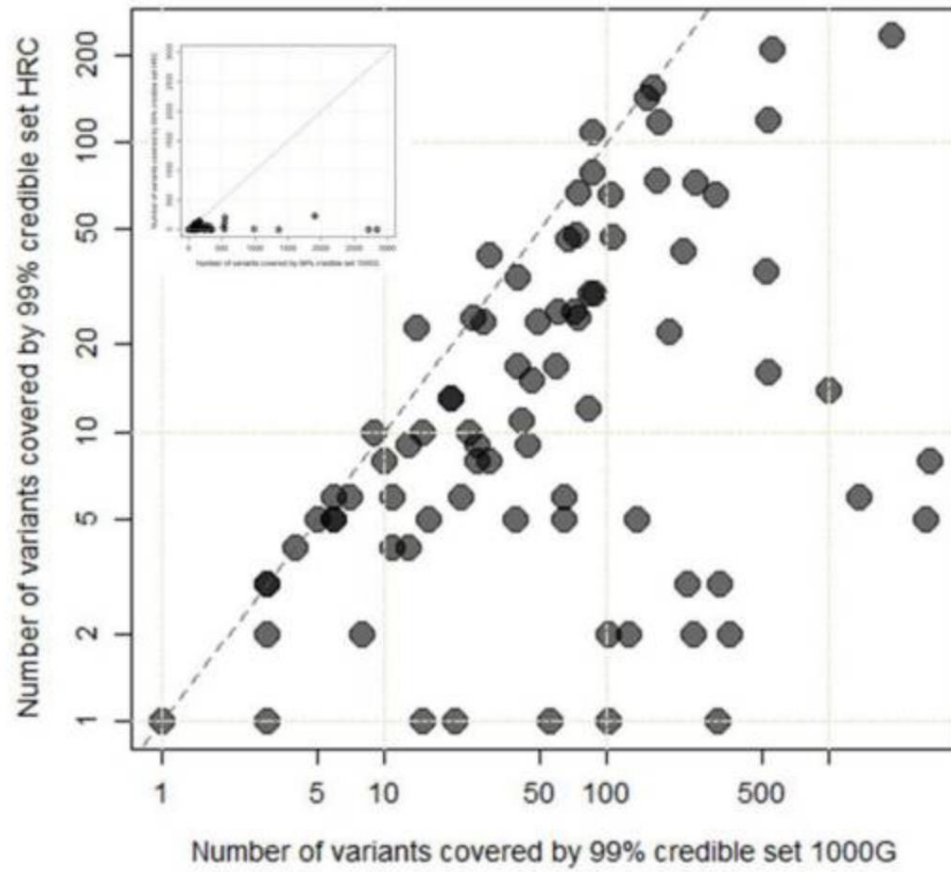
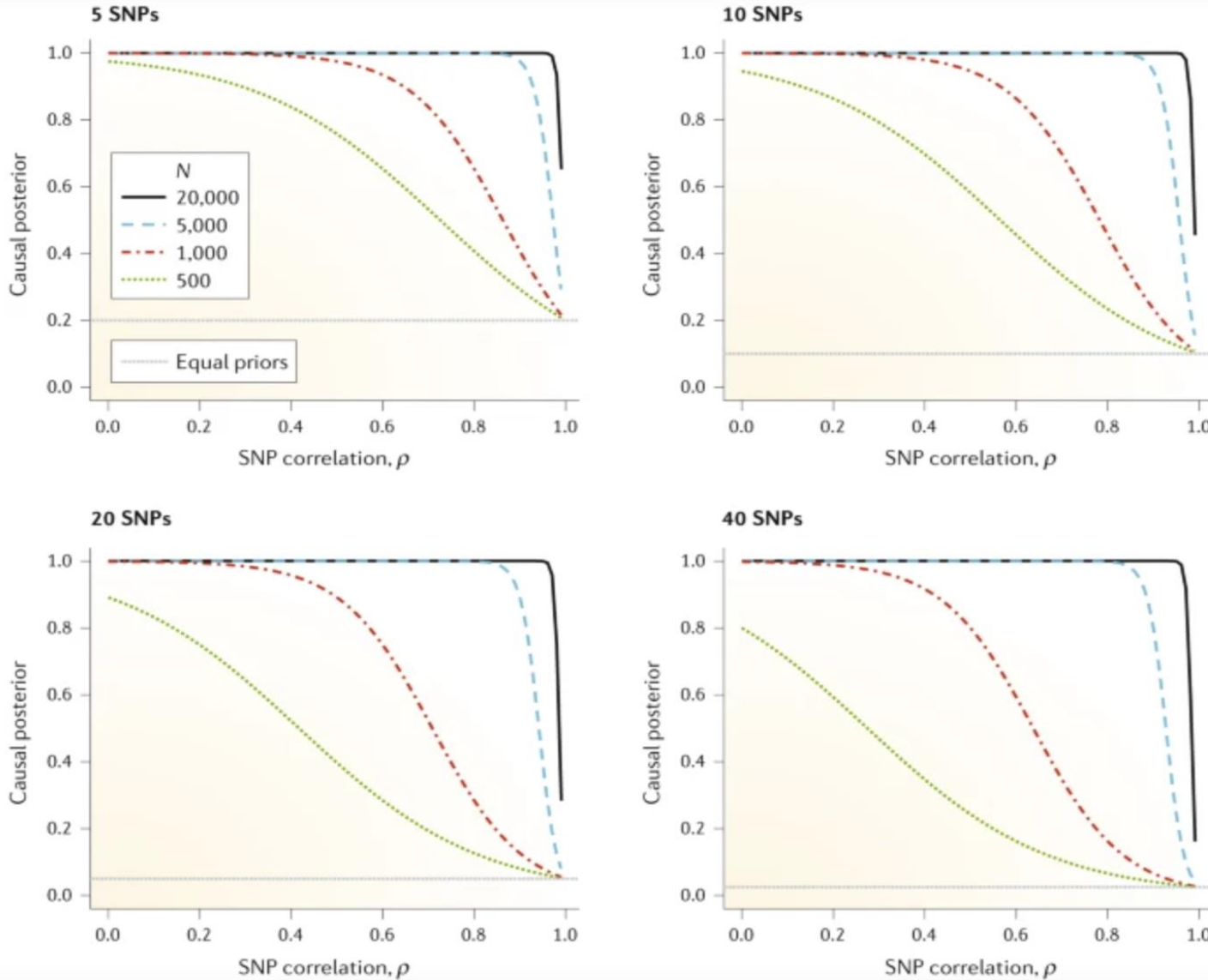


Image from: Mahajan et al (2018); *Nat. Gen.*

What determines fine-mapping resolution?



- **LD structure in the region**
 - Many SNPs in high LD with causal variants PIP would be almost same for all the variants and CS would contain many SNPs
- **Sample size of initial GWAS**
 - The study needs to be well powered to be informative
- **Effect size**
 - High effect size causal effects are easier to distinguish

Fine mapping with multiple causal variants

Several effects can overlap within one locus

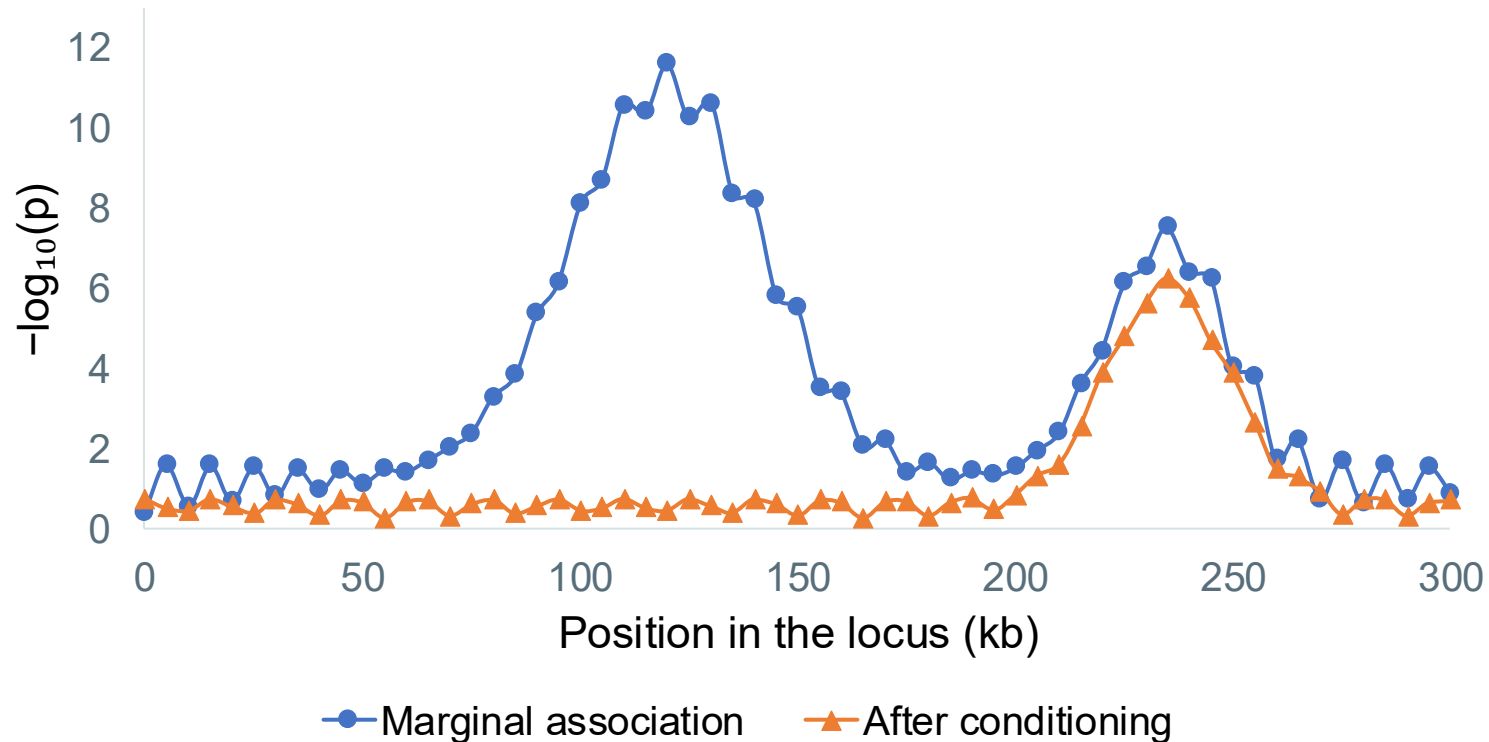
Why allow more than one causal variant?

- **Single causal variant might not be optimal** model for many scenarios especially for molecular phenotypes
- Multiple causal variants methods developed to incorporate a **wider range of causal scenarios** (might not be exhaustive)
- How to determine if the association signal at a locus is driven by a single causal variant or multiple causal variant?
Conditional Regression methods

Stepwise conditional regression

- **Step 1:** Take the sentinel SNP in the region and initialize the set S_c
- **Step 2:** Rerun the GWAS analysis conditioning on all currently selected SNPs
If SNP $_x$ is significant then, $S_c = S_c \cup \text{SNP}_x$
If no SNP is significant then stop
- Repeat Step 2 until no additional SNPs are significant
- Ideally S_c should give tag SNPs that have conditionally distinct associations with the outcome

Conditional analysis searches for residual signals



First, identify a strong association.

Include that variant as a covariate.

Ask whether another association remains.

A selected tag can absorb only part of a signal, or blur another one.

Limits of conditional regression

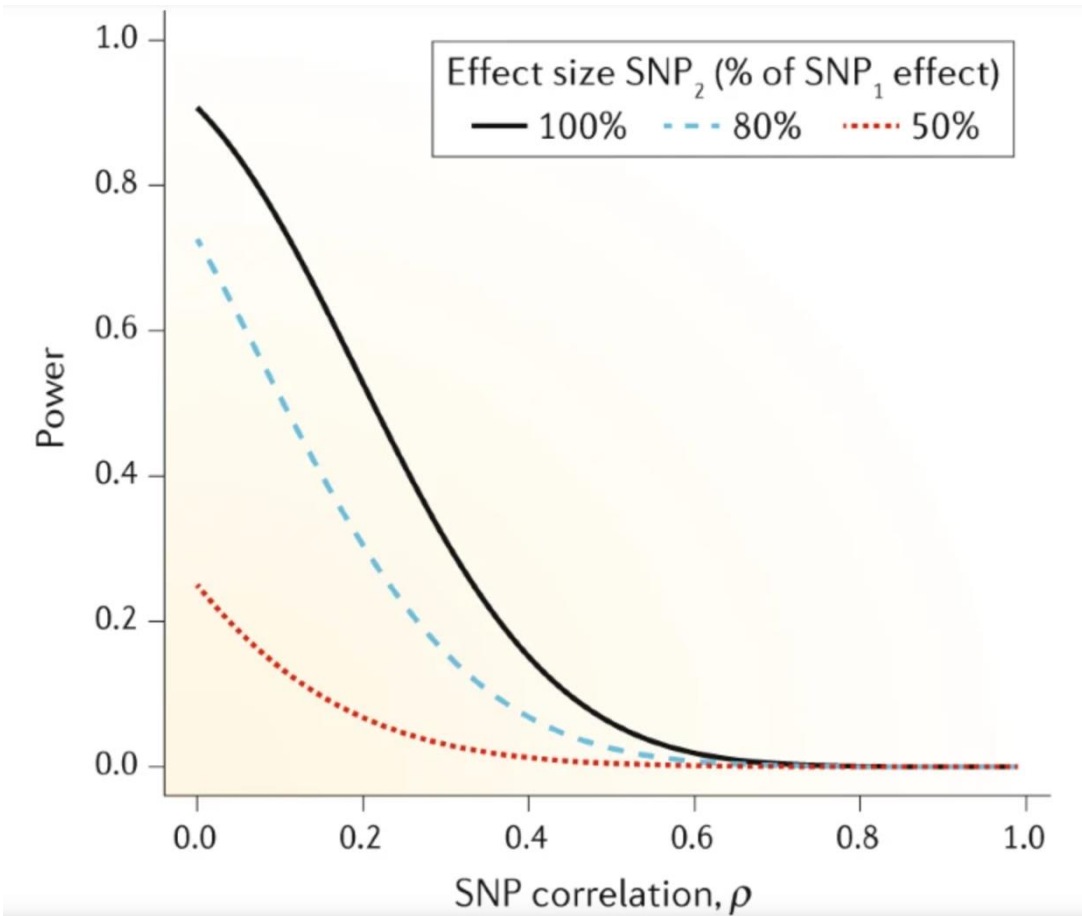


Image from: Schaid et al (2018); *Nat. Rev. Genetics*

Low statistical power with **increasing number of steps** (need to control for sequential testing)

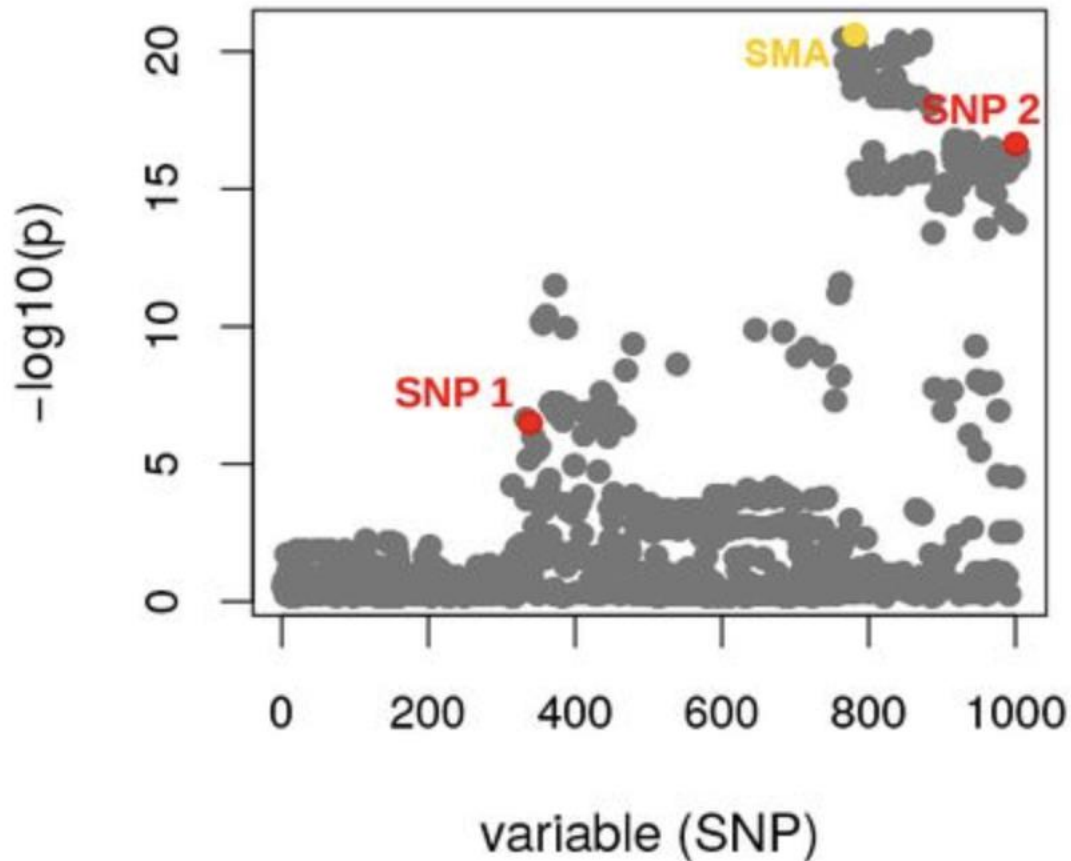
Many SNPs + using lenient cutoffs to include SNPs at each step can make selection unstable.

Simulation (Schaid et al.) with primary signal heritability 1% and sample size such that the power to detect the primary signal is 90%

Very low power with increasing LD with the primary signal

Very low power with weaker secondary signal

Complex LD makes signals hard to separate



Complicated LD patterns in the region and between the causal variants

Splitting the region into sub-regions driven by single causal variant might be difficult

Comprehensive and newer methods jointly model the data rather than splitting into sub-regions

Probabilities for combinations of variants

Possible causal configuration	Posterior probability
A only	0.10
B only	0.10
A and C	0.50
B and C	0.30

$$\text{PIP}(A) = 0.10 + 0.50 = 0.60$$

$$\text{PIP}(B) = 0.10 + 0.30 = 0.40$$

$$\text{PIP}(C) = 0.50 + 0.30 = 0.80$$

Variant PIPs sum to 1.80 here, the posterior expected number of effects.

The space of possible causal configurations

Each SNP can be included or excluded

m variants give 2^m possible configurations

20 variants: 1,048,576 configurations

Methods restrict, search, or approximate this space.
Priors often favor a small number of non-zero effects.

Summary-data methods: strengths and limitations

Method	Useful feature	Main limitation / check
Single-effect ABF	Fast, transparent, no LD matrix needed	Exactly one causal variant assumed
SuSiE-RSS	Efficient multiple effects and CSs	LD mismatch, L, convergence and CS purity
FINEMAP	Searches joint configurations	Search settings and maximum causal count
PAINTOR	LD + learned annotation priors	Annotation model and configuration complexity
CAVIAR / DAP-G	Alternative configuration-based inference	Bounds / approximations and computational cost

All require careful assumptions. No method guarantees a small, correct set.

SuSiE represents a region as a sum of effects

Models joint effects as a sum of L single-effect components.

Modeled effect	Where its probability concentrates
Effect 1	A, B, or C
Effect 2	F or G
Unused effects	May shrink toward zero

Each component considers one effect at an uncertain variant.

SuSiE repeatedly updates components while accounting for the others.

L (chosen by the analyst) sets an upper bound on modeled effects.

Each component can yield a credible set of alternative variants.

Multiple signals produce separate credible sets

Effect	Candidate probabilities	95% credible set
1	A 0.55, B 0.43, others 0.02	{A, B}: 0.98 total
2	F 0.45, G 0.30, H 0.22, others 0.03	{F, G, H}: 0.97 total

One CS describes uncertainty about one modeled effect.

SuSiE builds CSs from component probabilities.

Overall SNP PIPs can sum to more than one across the locus.

Fine mapping with genomic annotations

External evidence changes prior probabilities

Functional annotations can inform the prior

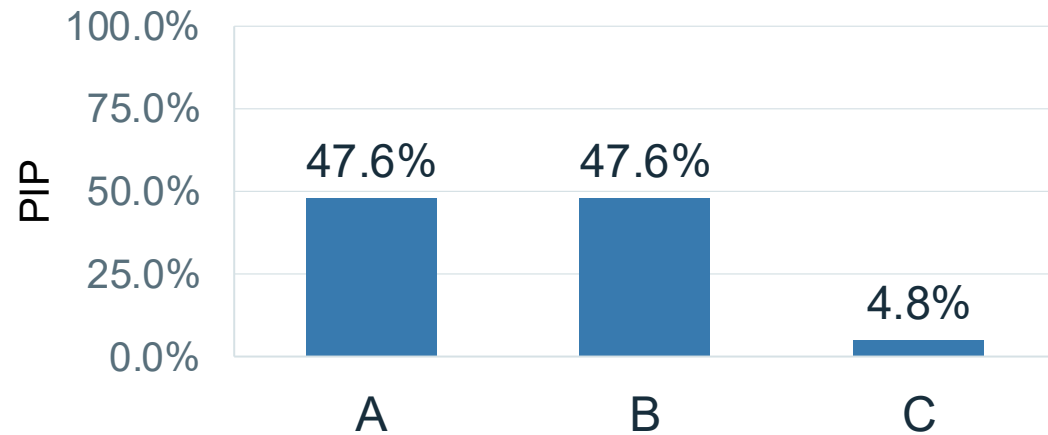
- From genomic studies we know certain types of variants are **enriched among candidate effect variants**
- Higher chances of variants with certain **genomic features** to be causal
- **Adjust the prior** to incorporate such genomic features of annotation
- These are referred to ordinarily as **“functional” prior** (priors from functional annotations)

What functional annotations contribute

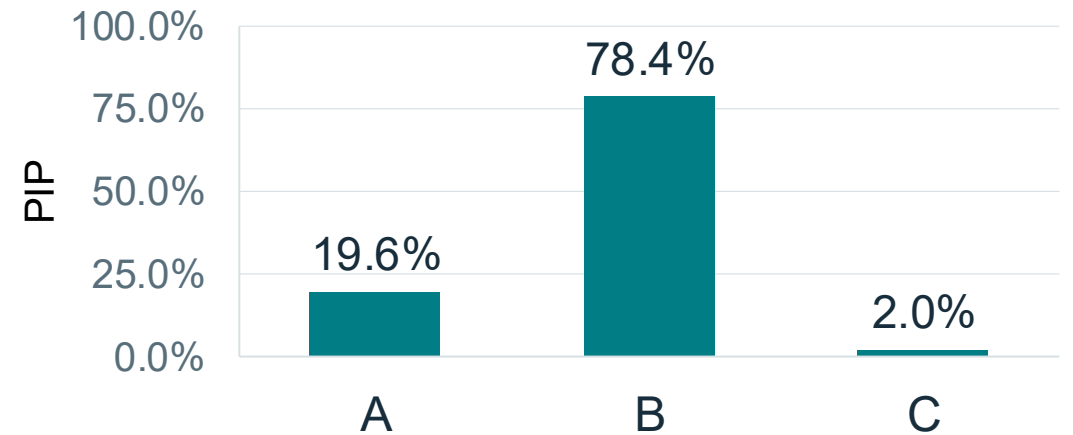
Annotation	Why it may inform a prior	Biological causality still unresolved
Coding consequence	Changes protein sequence or splicing	Does that change affect this trait?
Relevant enhancer / open chromatin	Variant lies in active regulatory DNA	Which allele, target gene, and cell state?
Binding motif / regulatory assay	Suggests a molecular effect	Does the mechanism act in the relevant context?

An annotation can separate similar candidates

No additional information

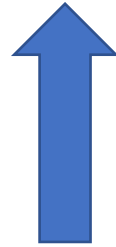


Functional information upweights B



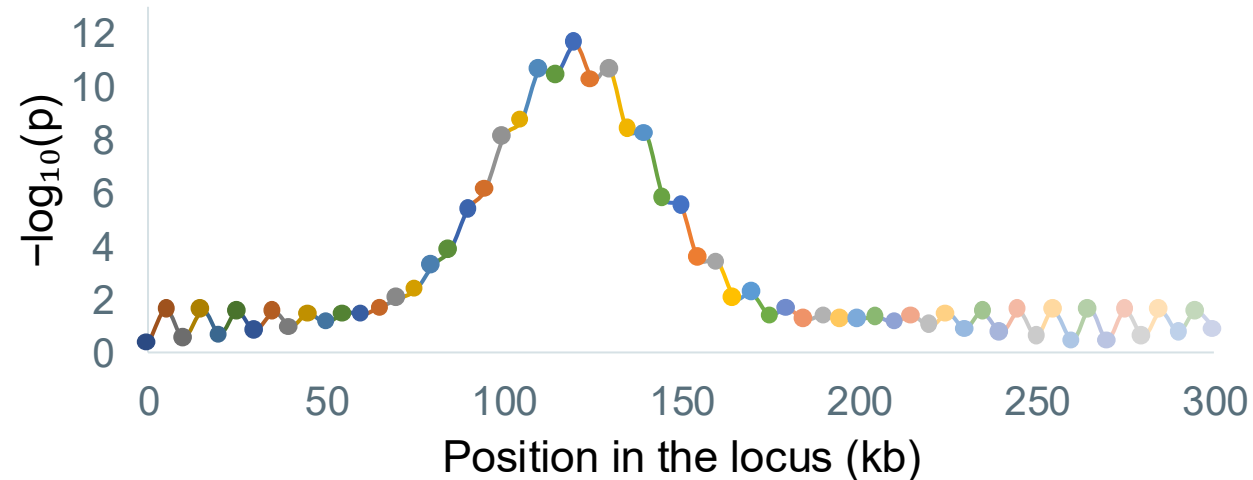
Association evidence and functional priors

True Causal
Status

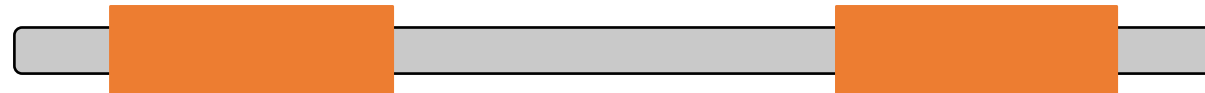


Infer which SNP(s) are causal in the locus
after adding biological/genomic information

Observed
association
patterns in GWAS



Genomic Features



Methods that incorporate functional priors

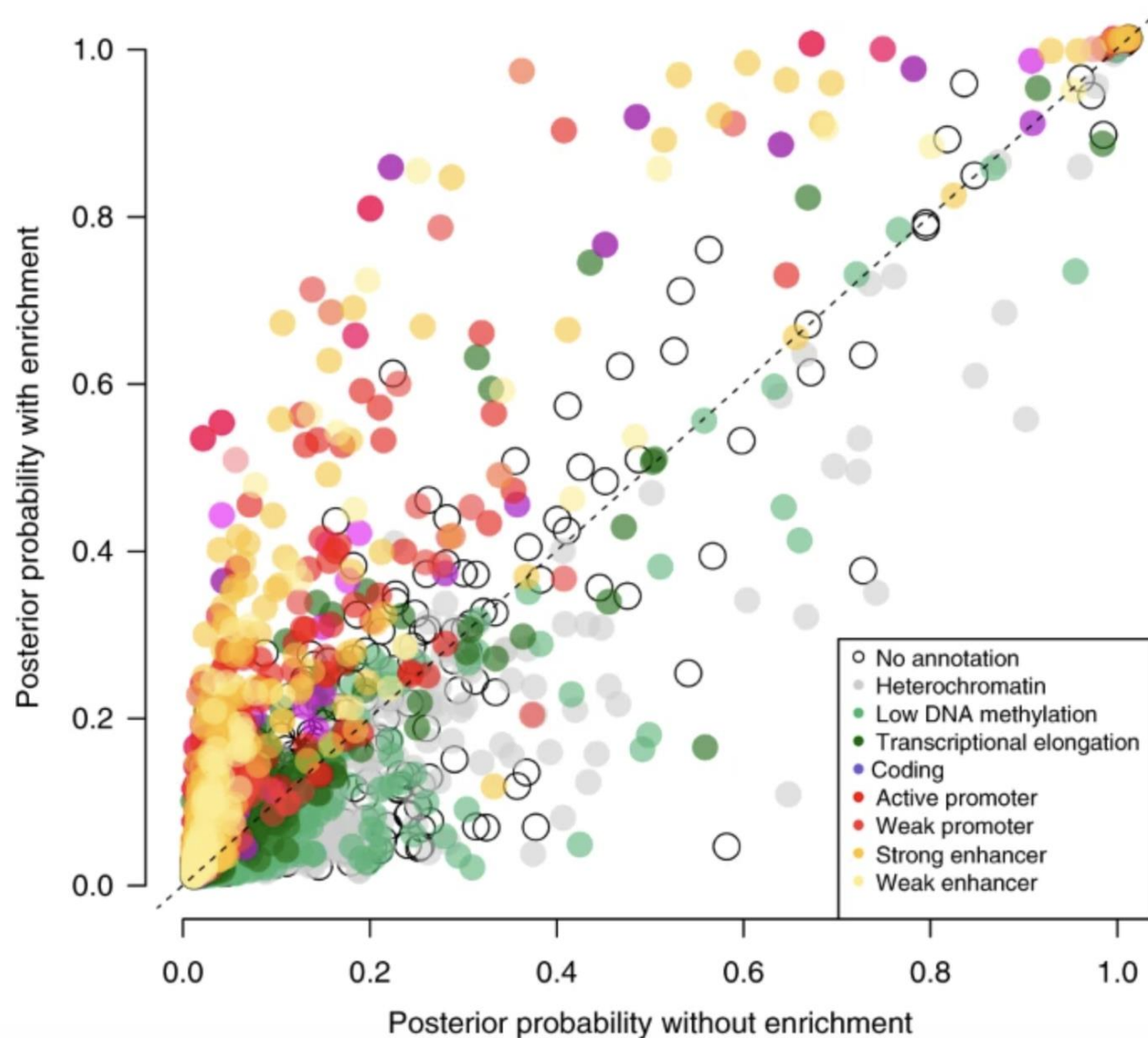
Method	How annotations enter	Consideration
fGWAS	Learns enrichment across genomic blocks	Classic one-signal block model
PAINTOR	Learns annotation-dependent priors with LD	Choice of annotations and causal-count setting
CAVIARBF	Bayes factors with annotation-informed priors	Prior fitting and model-search assumptions
PolyFun + SuSiE / FINEMAP	Polygenic functional enrichment supplies priors	Population match and avoiding overfitting

Annotation sources

Coding predictions: CADD, SIFT, PolyPhen

Regulatory evidence: ENCODE, Roadmap Epigenomics, RegulomeDB

Functional priors can change variant probabilities

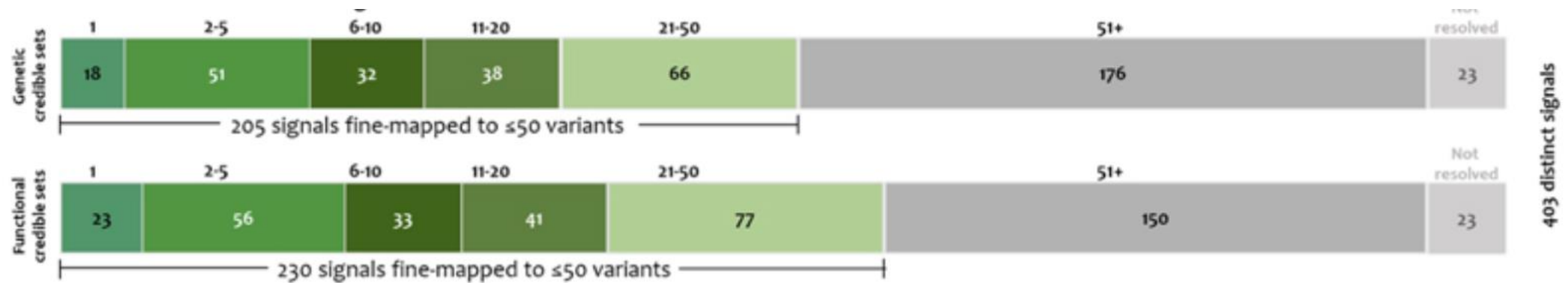


Analysis from Mahajan et al (2018):
GWAS analysis with 74,124 Type 2
diabetes cases and 824,006 controls

Colors signify SNPs different functional
category (near TSS, promoter, enhancer
etc.) in pancreatic islet

Image from: Mahajan et al (2018); *Nat. Gen.*

Functional priors can improve credible-set resolution



Incorporating functional annotations as priors can improve resolution of credible sets: Smaller credible sets

Provides much granular information in comparison to noninformative prior

Functionally informed fine mapping across traits

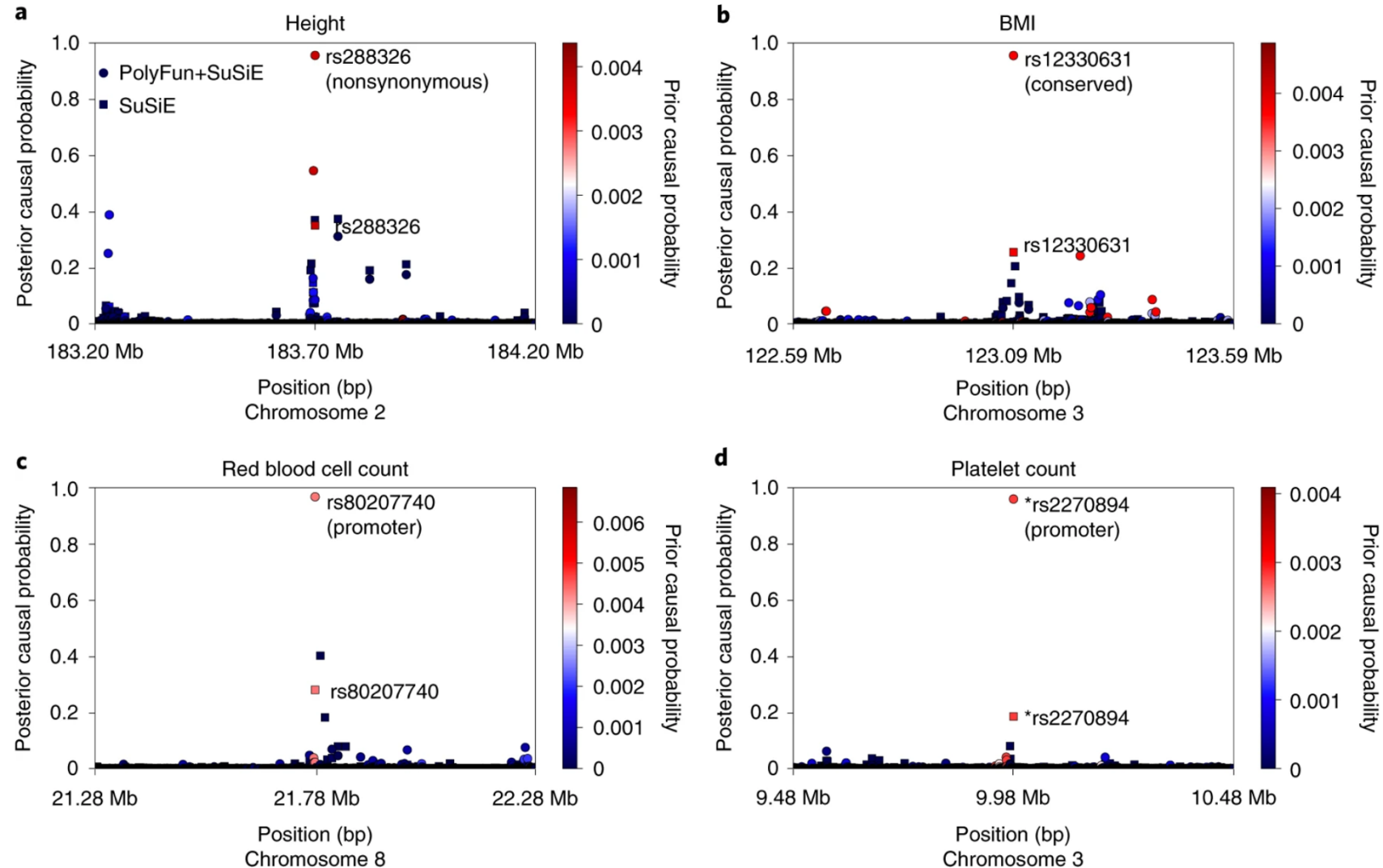


Image from: Weissbrod et al (2020); *Nat. Gen.*

Considerations for functional priors

- **Genomic annotations can be specific** to tissue, cell type and cell developmental stages
- Incorporating **misinformed prior** might blur inference and introduce noise
- Incorporating **too many annotations** might not be beneficial always
- Choice of the prior causal probability (PolyFun addresses this)

Multi-ancestry fine mapping

Different LD patterns can narrow the candidates

Different LD patterns provide different clues

B is the only causal variant shared across ancestries/populations

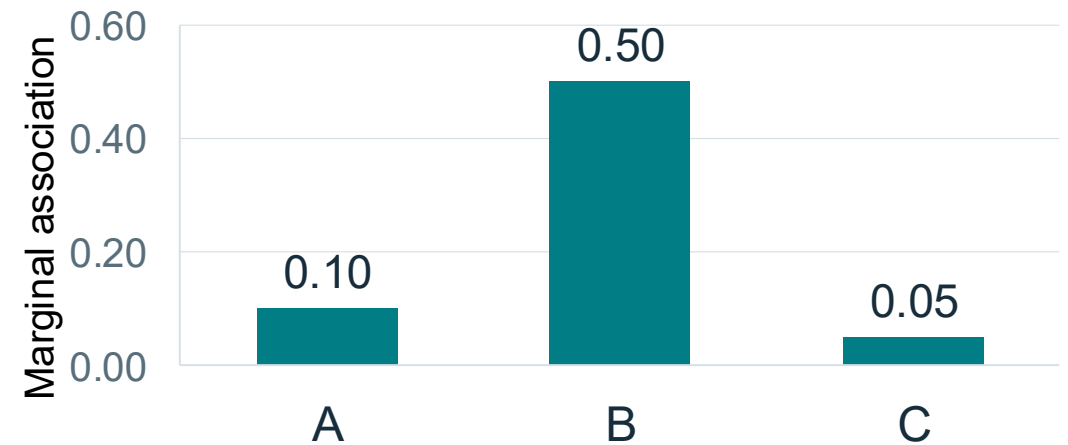
EUR-like toy

$$r_{AB} = 0.98, r_{BC} = 0.20$$



AFR-like toy

$$r_{AB} = 0.20, r_{BC} = 0.10$$



Combining complementary evidence

Candidate	EUR-like BF	AFR-like BF	Product	Joint PIP
A	10	2	20	0.0384
B	10	50	500	0.9597
C	1	1	1	0.0019

Study 1 alone: A and B each have PIP 0.476.

Combined: B exceeds 0.95 and forms a singleton 95% credible set.

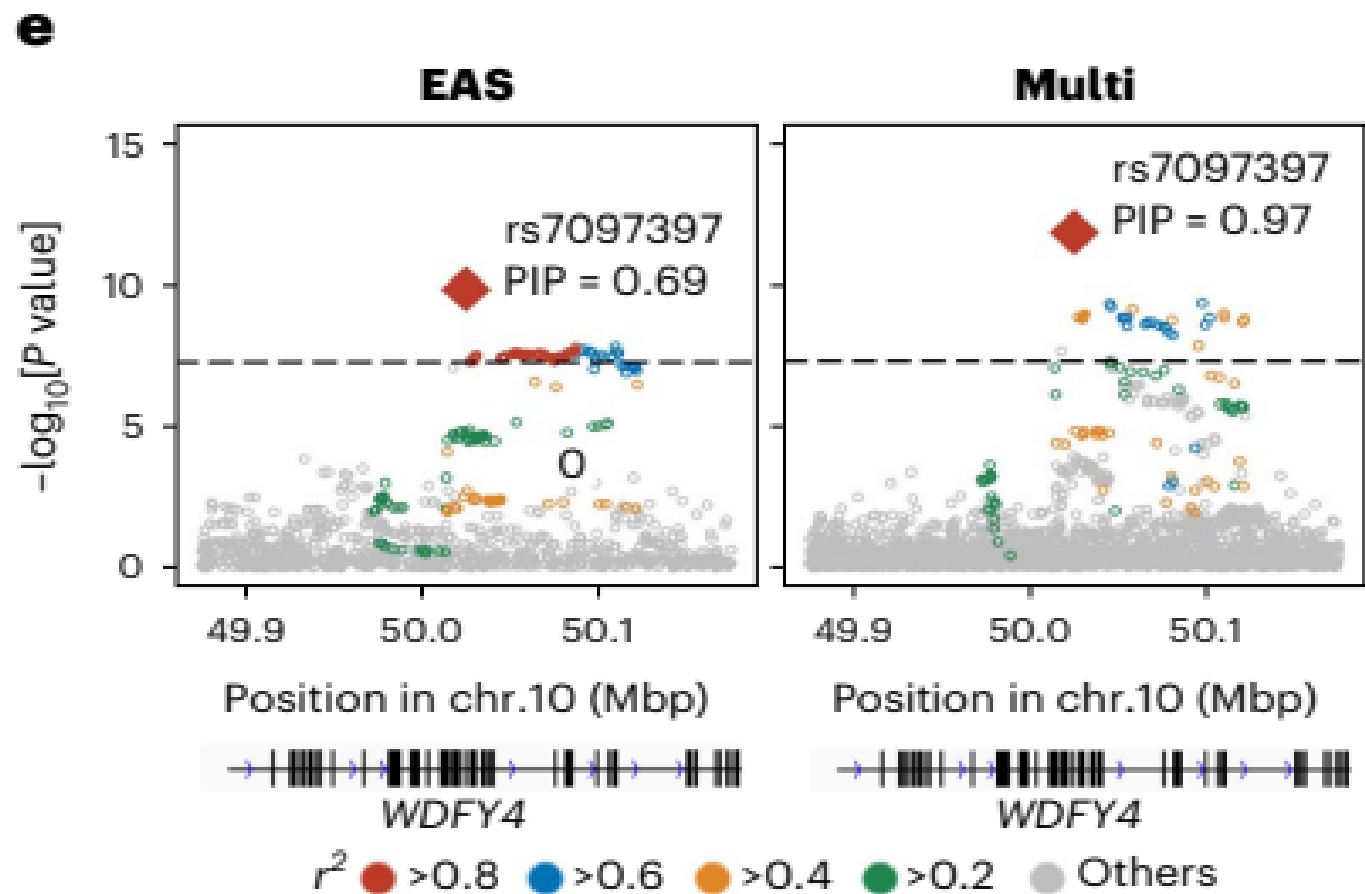
Multi-ancestry analysis with summary statistics

Population	Association evidence	Matched correlation evidence
EUR	Effect size, SE, N	R_{EUR} in the same SNP / allele order
AFR	β Effect size, SE, N	R_{AFR} in the same SNP / allele order
Additional groups	Their own association summaries	Their own LD matrices

Joint evidence = combine population-specific likelihoods

Shared candidate locations, potentially different effect sizes
Joint PIPs and credible sets under the chosen sharing model

Example: rheumatoid arthritis at WDFY4



95% credible-set size

Analysis	Variants
EUR	6,391
EAS	64
Multi-ancestry	1

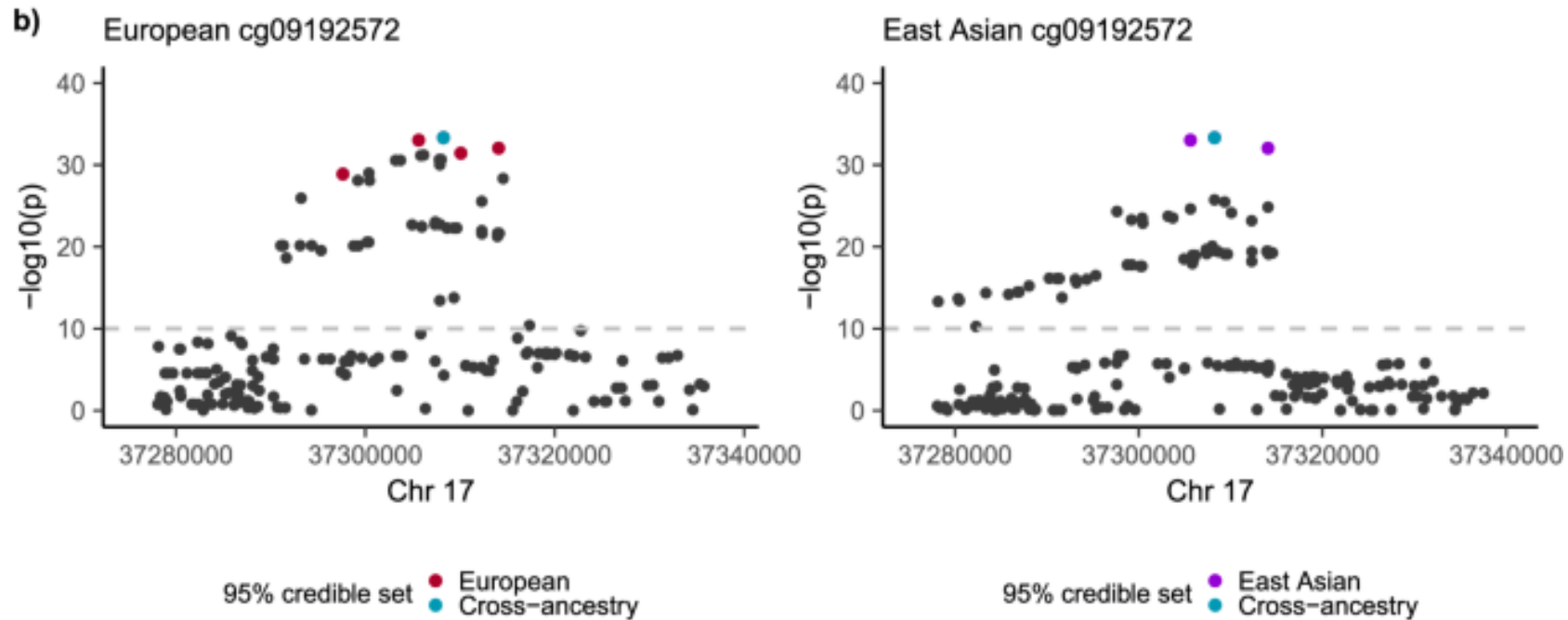
rs7097397 (WDFY4 missense)
PIP: 0.69 EAS, 0.97 combined

Gain reflects LD, allele frequency, and sample size.

Example: methylation at cg09192572

SuSiEx combined mQTL summary statistics with EUR and EAS LD.

95% credible sets



EUR: 5 SNPs

EAS: 3 SNPs

Joint: 1 SNP

Multi-ancestry methods: strengths and limitations

Method	Main strength	Main assumption / trade-off
SuSiEx	Flexible number of populations Fast summary-data inference	Shared variant locations Independent cross-population effect priors
MultiSuSiE	Correlated effects can borrow strength Individual or summary data	Shared locations Effect covariance / prior specification
SuShiE	Learns effect sharing Molecular QTL and prediction workflow	Shared cis-effect locations Covariance estimation needs information
MESuSiE	Models shared and ancestry-specific causal patterns	More flexible sharing model Potentially greater computational burden

Discussion: when will another ancestry help?

A large EUR GWAS + a smaller AFR GWAS: when does resolution improve?

Consideration	What to think about
LD and information	Are close EUR tags less correlated in the second dataset?
Frequency and coverage	Are candidate variants polymorphic, well imputed, and measured?
Effect sharing	Same phenotype and plausible shared effects? Heterogeneity?
LD representation	Matched panels, admixed cohorts, local ancestry, sample overlap?
Robustness	Compare per-ancestry results and sensitivity to priors / LD.

More diverse data can help. Improvement is not guaranteed at every locus.

Practical fine mapping: preparing a GWAS locus

Step	Checks and inputs
1. Define the locus	Include correlated variants and nearby signals. Retain non-significant SNPs.
2. Harmonize variants	Genome build, chr:pos:ref:alt, alleles / strand, duplicates, imputation INFO.
3. Assemble statistics	Effect sizes + SE (or signed z), N, allele frequency, trait type and case / control counts.
4. Match LD	Signed R, identical variant order and effect-allele coding, matched population. Best: in-sample LD
5. Specify the model	Maximum effects L / K, coverage level, prior, covariates and LD provenance.

Practical fine mapping: LD and diagnostic checks

LD source	Use and limitation
GWAS participants	Preferred. Match samples, dosages, covariate treatment, and allele coding.
Closely matched reference cohort	Use enough participants and good coverage. Check frequency / LD mismatch.
1000 Genomes	Public genotypes to calculate LD. Small ancestry subsets can be noisy.
Pan-UK Biobank	Public ancestry-specific LD matrices. Check build, variant map and cohort match.

Compute signed LD

Before interpretation: z-LD consistency, convergence, CS purity, and sensitivity to LD, region boundaries, and the number of effects.

Fine mapping: the main messages

Question

Which variants explain the locus under a joint model?

Uncertainty

PIPs and credible sets distinguish prioritization from certainty.

Extra information

Annotations and ancestry-specific LD can separate close candidates.

Next step

Check assumptions, then combine with molecular and experimental evidence.

Colocalization

Do two traits share a causal genetic variant?

Shared variants and shared mechanisms

Colocalization asks whether two traits share a causal variant.

Shared variant	Different variants in LD
Variant A affects both traits	Variant A affects trait 1 Variant B affects trait 2
Compatible with a shared genetic signal	Nearby peaks can look similar
Mediation or separate biological pathways can both produce this pattern	Fine mapping helps distinguish linked effects

Shared variant $\neq$ proof that one trait causes the other

The five hypotheses in the original coloc model

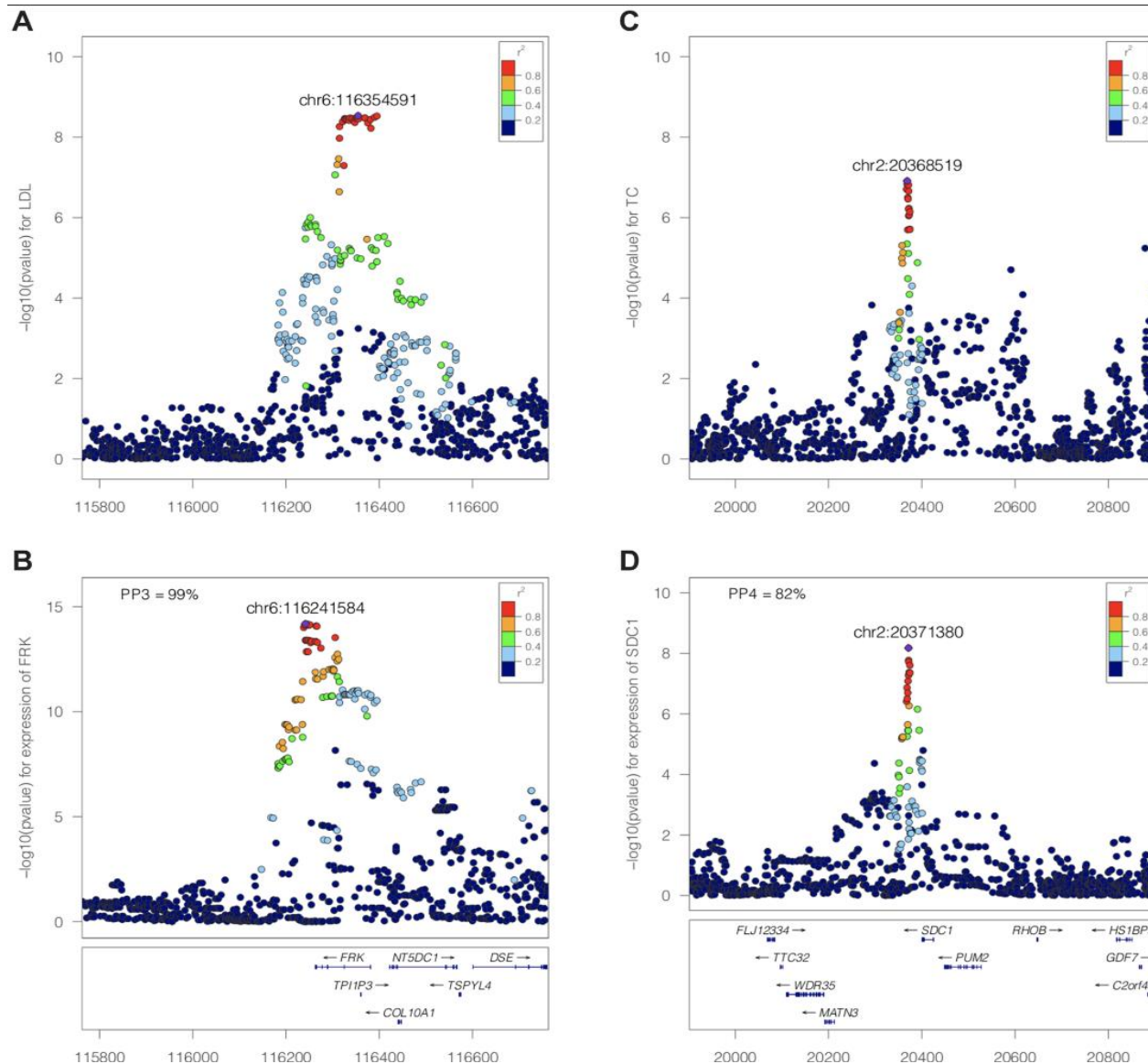
Hypothesis	Interpretation
H0	Neither trait associated
H1	Trait 1 associated, trait 2 not associated
H2	Trait 2 associated, trait 1 not associated
H3	Both associated, different causal variants
H4	Both associated, one shared causal variant

Original coloc: at most one causal variant per trait in the region

Region-level and variant-level outputs

Region / signal level	coloc: PPH0–PPH4 summarize support for each hypothesis.
Variant identity	coloc: SNP.PP.H4 ranks shared candidates conditional on H4.
Variant colocalization	eCAVIAR: CLPP uses variant fine-mapping probabilities under its model.
Interpretation	PPH4 and CLPP answer different questions. Their cutoffs are not interchangeable.

Two loci with different colocalization conclusions



A–B: LDL and liver FRK expression
Evidence for different variants
PPH3 $\approx$ 99%

C–D: total cholesterol and liver SDC1 expression
Evidence supporting a shared variant
PPH4 $\approx$ 82%

Image from: Giambartolomei et al (2014); *Plos. Gen.*

Interpreting a colocalization result

Power

Low PPH4 can reflect weak evidence for one or both traits.

Sharing vs linkage

Compare H4 with H3, while reporting all five probabilities.

Model and data

Check multiple signals, variant coverage, LD, and sample overlap.

Sensitivity

Examine plausible priors and region boundaries before a biological claim.

Pop quiz 2: what does colocalization tell us?

Disease and a liver eQTL have $PPH4 = 0.95$.

What can we conclude?

- A. Gene expression definitely mediates disease risk.
- B. The data support a shared variant under the model.

Colocalization with multiple signals

Fine-map each trait, then compare signal pairs

	eQTL effect 1	eQTL effect 2
Disease effect 1	Shared signal?	Shared signal?
Disease effect 2	Shared signal?	Shared signal?

coloc.susie uses component-level evidence from SuSiE fits.

Overlapping credible sets alone are not a colocalization test.

Colocalization software and required inputs

Method	Inputs	Output / limitation
coloc.abf	Summary effects + variances Trait-specific metadata	PPH0–PPH4 At most one effect per trait
coloc.susie	Summary data + matched signed LD (or existing SuSiE fits)	Signal-pair PPH0–PPH4 Depends on fine-mapping quality
eCAVIAR	Signed z scores + LD for each trait	Variant CLPP Multiple effects with a chosen bound

Software links and detailed references are in the notes.

Questions and discussion

Which claim does your evidence support?

An associated locus

A prioritized variant

A shared signal across traits

A demonstrated biological mechanism

Thank you



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